O-RAN.WG7.NES.0-v03.00

Technical Report

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O-RAN Work Group 7 (White-box Hardware Workgroup)

Network Energy Savings Procedures and Performance Metrics

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Foreword

This Technical Report (TR) has been produced by O-RAN Alliance.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules [14] (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Executive summary

This technical report describes functional procedures for Network Energy Savings, with focus on the O-RU. Energy savings KPIs and estimated power reductions are also presented. The mapping of energy savings to existing M-Plane and C-Plane commands is discussed, with gaps noted (i.e., new command capabilities requiring development in O-RAN WG4.) In follow-on versions, there will be elaboration on Advanced Sleep modes, and details on O-RU hardware and firmware impact will also be covered.

1 Scope

The contents of the present document are subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

The present document describes Network Energy Savings methods and requirements associated mainly with the O-RU. The O-DU or CU are referenced with respect to commands from these entities, or data exchanged between O-RU and O-DU pertaining to NES. Both TDD and FDD deployments are covered as well as FR1 and FR2.

The NES mechanisms described herein do not assume a shared O-RU architecture. Such an arrangement adds complexity to the low power control, as multiple O-DUs or O-CUs would be sharing a single O-RU. NES requirements and functionality in a shared O-RU scenario is a topic for future study.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ORAN-WG7.DSC.0-V02.00 Technical Specification, 'Deployment Scenarios and Base Station Classes for White Box Hardware'.

[2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[3] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception". (Rel-16.3.0)

[4] ORAN-WG4.CUS.0-v04.00 Technical Specification, 'O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification'.

[5] ETSI ES 202 706-1 V1.7.0 (2022-06), "Metrics and measurement method for energy efficiency of wireless access network equipment; Part 1: Power consumption - static measurement method"

[6] O-RAN Architecture Description, O-RAN-WG1-O-RAN Architecture Description - v01.00.00

[7] O-RAN WG6, Cloud Architecture and Deployment Scenarios for O-RAN Virtualized RAN.

[8] ORAN-WG4.MP.0-v04.00 Technical Specification, WG4 O-RAN Management Plane Specification'.

[9] O-RAN TIFG O-RAN End-to-End System Testing Framework v1.0.

[10] O-RAN White Box Hardware Working Group Outdoor Macrocell Hardware Reference Design Specification

[11] O-RAN.WG1.Network-Energy-Savings-Technical-Report-R003-v01.00

[12] 3GPP TS 28.310 Technical Specification "Group Services and System Aspects; Management and orchestration; Energy efficiency of 5G", V17.4.0

[13] ETSI ES 203 228, "Environmental Engineering (EE); Assessment of mobile network energy efficiency", V1.3.1

[14] O-RAN Drafting rules: https://oranalliance.atlassian.net/wiki/download/attachments/2203189326/O-RAN.TSC.Drafting-Rules.0-v01.00.docx?api=v2

[15]

ETSI TS 138 141-1, 5G; NR; Base Station (BS) conformance testing Part 1: Conducted conformance testing

[16] https://oranalliance.atlassian.net/wiki/download/attachments/2336817208/O-RAN.WG1.Network-Energy-Savings-Technical-Report-R003-v02.00.pdf?api=v2

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[1] https://telecominfraproject.com/wp-content/uploads/OpenRANTechnicalPriorityDocument-Rel3.xlsx.

[2] M. Azizi et al, "Energy Router: A Sustainable Solution for Future Residential Buildings" IEEE Power Electronics Magazine, March 2025.

[3] IEEE 1547, https://standards.ieee.org/ieee/1547/5915/

[4] SunSpec Device Information Model Specification, https://sunspec.org/specifications/#sunspec-modbus-for2b84-f1df

[5] N. Deltimple et al, "Dynamic Biasing Techniques for RF Power Amplifier Linearity and Efficiency Improvement", IEEE International Conference on Integrated Circuit Design and Technology, June 2010

[6] EPC application note AN028, "Envelope Tracking Power Supply for Cell Phone Base Stations"

[7] C. Hsia, "High-Efficiency Wideband Envelope-Tracking Power Amplifier Module with GaN PA for 5G NR Base-Station Applications", 2023 IEEE 66th International Midwest Symposium on Circuits and Systems

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

Active Mode reference power consumption: this refers to the O-RU (or O-RU/DU combo box) power consumption while user PRBs are being transmitted or received. Unless otherwise noted, for a TDD configuration we will define this to be the average O-RU power consumption for a continuous 75/25 DL/UL duty cycle, with allowance for PDCCH and PUCCH traffic as well as required symbol gaps. By default, it assumes that all available RF signal chains are enabled, for a given band, at the max bandwidth supported by the O-RU.

Advanced Sleep Mode reference power consumption: this refers to the O-RU or (O-RU/DU combo box) power consumption when configured in one of the four defined sleep modes SM0-SM3. Note that a given O-RU may support only a subset of sleep modes SM0-SM3, or none at all. By default, the ASM reference power consumption is expressed in Watts, but a relative % reduction of power, vs the Active Mode reference power consumption above, is also acceptable.

Beam: main lobe of a radiation pattern from an AAS BS

NOTE: For certain AAS BS antenna array, there may be more than one beam.

Beamwidth: In a radiation-pattern cut containing the direction of the maximum of a lobe, the angle between the two directions in which the radiation intensity is one-half the maximum value, also called as half power beamwidth.

Carrier aggregation: aggregation of two or more NR or E-UTRA component carriers in order to support wider transmission bandwidths. Within a band, component carriers can be adjacent (contiguous) or non-contiguous. If multiple component carriers are sent in different bands, this corresponds to inter-band carrier aggregation.

Carrier Frequency: Center frequency of the cell.

F1 interface: The standard interface between an O-CU and an $O\text{-}DU_x$ specified in [8].

Integrated architecture: An architecture wherein the $O\text{-}RU_x$ and $O\text{-}DU_x$ are implemented on one platform. Each $O\text{-}RU_x$ and RF frontend are associated with one $O\text{-}DU_x$ and are connected to O-CU via F1 interface

Split architecture: An architecture wherein the $O\text{-}RU_x$ and $O\text{-}DU_x$ are physically separated from one another. A switch may aggregate multiple $O\text{-}RU_x$ (s) to one $O\text{-}DU_x$. The $O\text{-}DU_x$, switch and $O\text{-}RU_x$ (s) are connected by the fronthaul interface as defined in [4].

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

Occupied Bandwidth (OBW): It refers to the bandwidth occupied by the base station when operated, defined by the sum of the active bandwidths of the band allocation(s) operated. As defined by 3GPP TS 34.121 section 5.8, occupied bandwidth is the bandwidth containing 99% of the total integrated power of the transmitted spectrum, centered on the assigned channel frequency. The bandwidth between the 0.5% power frequency points is the occupied bandwidth.

Instantaneous Bandwidth (IBW): It refers to the bandwidth in which all frequency components can be simultaneously analyzed. It is defined by the frequency boundaries of the operating band(s).

Frequency Range: It refers to bandwidth defined by the frequency range within which the Base Station can be operated, defined by the band-pass filter of the BS, e.g., 3.4 – 3.8 GHz (400 MHz)

gNB: A RAN node providing NR user plane and control plane protocol terminations towards the UE, and connected via the NG interface to the 5GC

Frequency Band: A designated frequency range for the operation of the base station and the UE radios. 5G NR frequency bands are divided into two different frequency ranges: Frequency Range 1 (FR1) that mainly includes sub-6GHz frequency bands, some of which are bands traditionally used by previous standards but has been extended to cover potential new spectrum offerings from 410MHz to 7125MHz; Frequency Range 2 (FR2) that includes the FR2-1 frequency bands from 24.25 GHz to 52.6 GHz, as well as the FR2-2 bands covering 52.6 to 71 GHz. Bands in this millimeter wave range have shorter range but higher available bandwidth than bands in the FR1.

Fronthaul Gateway (FHGW): A fronthaul gateway is a physical entity that is located between a distributed unit and one or more radio units where it distributes, aggregates, and/or converts fronthaul protocols between the distributed unit and multiple radio units.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

N_{RB}	Transmission bandwidth configuration, expressed in units of resource blocks (for E-UTRA and NR)

3.3 Abbreviations

For the purposes of the present document, the [following] abbreviations [given in ... and the following] apply:

Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x

3GPP	Third Generation Partnership Project
5G	Fifth Generation [Mobile Communications]
5GC	5G Core
AAS	Active Antenna System
ACLR	Adjacent Channel Leakage Ratio
ACS	Adjacent Channel Selectivity
ADC	Analog to Digital Converter
AFE	Analog Front End
AFU	Antenna Filter Unit
ASM	Advanced Sleep Mode(s)
BF	Beamforming
BMCA	Best Master Clock Algorithm
BPSK	Binary Phase Shift Keying
BS	Base Station
C2C	Chip to Chip Interface
CA	Carrier Aggregation
CAL	Calibration
CC	Component Carrier
CFR	Crest Factor Reduction
CLK	Clock
CPRI	Common Public Radio Interface
CU	Central Unit
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DFE	Digital Front End
DL	Downlink
DPD	Digital Pre-Distortion
DSP	Digital Signal Processor
DU	Distributed Unit
DUC	Digital Up Conversion
eCPRI	Enhanced Common Public Radio Interface
eNB	e-NodeB=LTE Base Station
gNB	g-NodeB=NR Base Station
EMC	ElectroMagnetic Compatibility

ES	Energy Savings
EVM	Error Vector Magnitude
FAPI	Functional Application Platform Interface
FEC	Forward Error Correction
FFT	Fast Fourier Transform
FH	Fronthaul
FHGW	Fronthaul Gateway
FHGWx→y	Fronthaul Gateway that can translate FH protocol from an O-DUx with split option x to an O-RUy with split option y, with currently available option 7-2→8.
FHM	Fronthaul gateway with no FH protocol translation, supporting an O-DU with split option x and an O-RU with split option x, with currently available options 6→6, 7-2→7.2, 8→8
FPGA	Field Programmable Gate Array
GbE	Gigabit Ethernet
GPP	General Purpose Processor
IEEE	Institute of Electrical and Electronics Engineers
IMD	InterModulation Distortion
IP	Internet Protocol
I/O	Input/Output
JTAG	Joint Test Action Group
L1	Layer 1, also referred to as PHY or Physical Layer
L2	Layer 2, also referred to as Data Link layer in OSI model
L3	Layer 3, also referred to as Network layer in OSI model
LLS	Lower Layer Split
LNA	Low Noise Amplifier
LTE	Long Term Evolution
MAC	Media Access Control
MIMO	Multiple Input Multiple Output
mMIMO	Massive Multiple Input Multiple Output
MCP	Multi-Chip Package
nFAPI	Network Functional Application Platform Interface
NES	Network Energy Savings
NC-CA	Non-contiguous Carrier Aggregation
NG	Next Generation
NR	New Radio
OAM	Operations, Administrations and Maintenance
O-CU	O-RAN Centralized Unit as defined by O-RAN
O-DU	O-RAN Distributed Unit as defined by O-RAN
O-DUx	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2 (as defined by WG4) or 8
O-RU	O-RAN Radio Unit as defined by O-RAN

O-RUx	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2 (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DUx
PA	Power Amplifier
PCIe	Peripheral Component Interface Express
PDCP	Packet Data Convergence Protocol
PHY	Physical Layer, also referred as L1
PLL	Phase Locked Loop
POE	Power over Ethernet
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RAN	Radio Access Network
RF	Radio Frequency
RFIC	Radio Frequency Integrated Circuit
RFSoC	Radio Frequency System on Chip
RLC	Radio Link Controller
RRC	Radio Resource Controller
RU	Radio Unit as defined by 3GPP
RX	Receiver
SDU	Service Data Unit
SFP	Small Form-factor Pluggable
SFP+	Small Form-factor Pluggable Transceiver
SoC	System on Chip
SPI	Serial Peripheral Interface
SSB	Synchronization Signal Block
ST	Section Type
TR	Technical Report
TS	Technical Specification
TX	Transmitter
UL	Uplink
VR	Voltage Regulator
WG	Working Group

4 Overview of Network Energy Savings Scenarios

The NES methods and requirements documented here can, in general, apply to all forms of O-RU deployment. Some specific Network Energy savings mechanisms may, however, not be relevant to certain O-RU architecture deployments.

4.1 Deployment Scenarios

An O-DU can interface to multiple O-RUs, as shown in the figure below. When Network Energy Savings modes are applied based on traffic reduction, as is often the case, both the O-DU and O-RU power consumption can be expected to decrease, but this document will focus on O-RU Network Energy Savings.

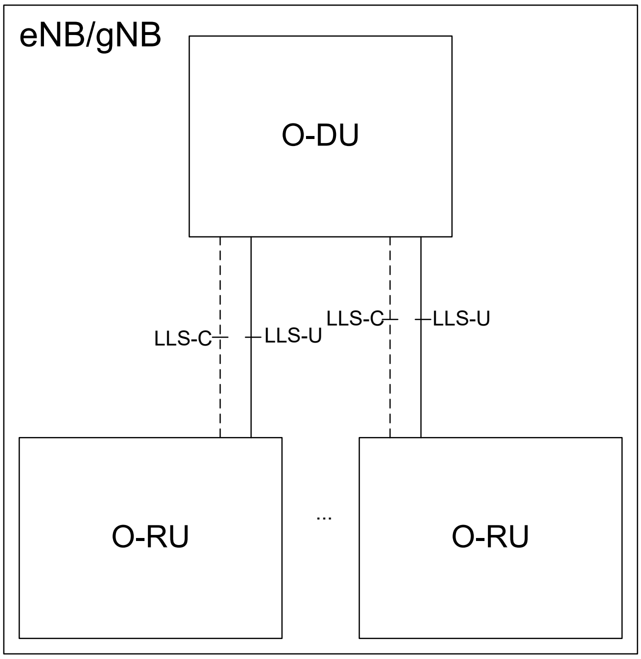

Figure 4-1 O-DU and O-RU architecture with interfaces

The document ORAN-WG7.OMAC.HRD-v01.00, which provides the Hardware Reference Design spec for a Split 7-2 Outdoor Macro Cell, gives the figure below in section 2.3.1. Generally speaking, all of the sub-blocks below are subject to Network Energy Savings as a function of traffic or operational scenarios.

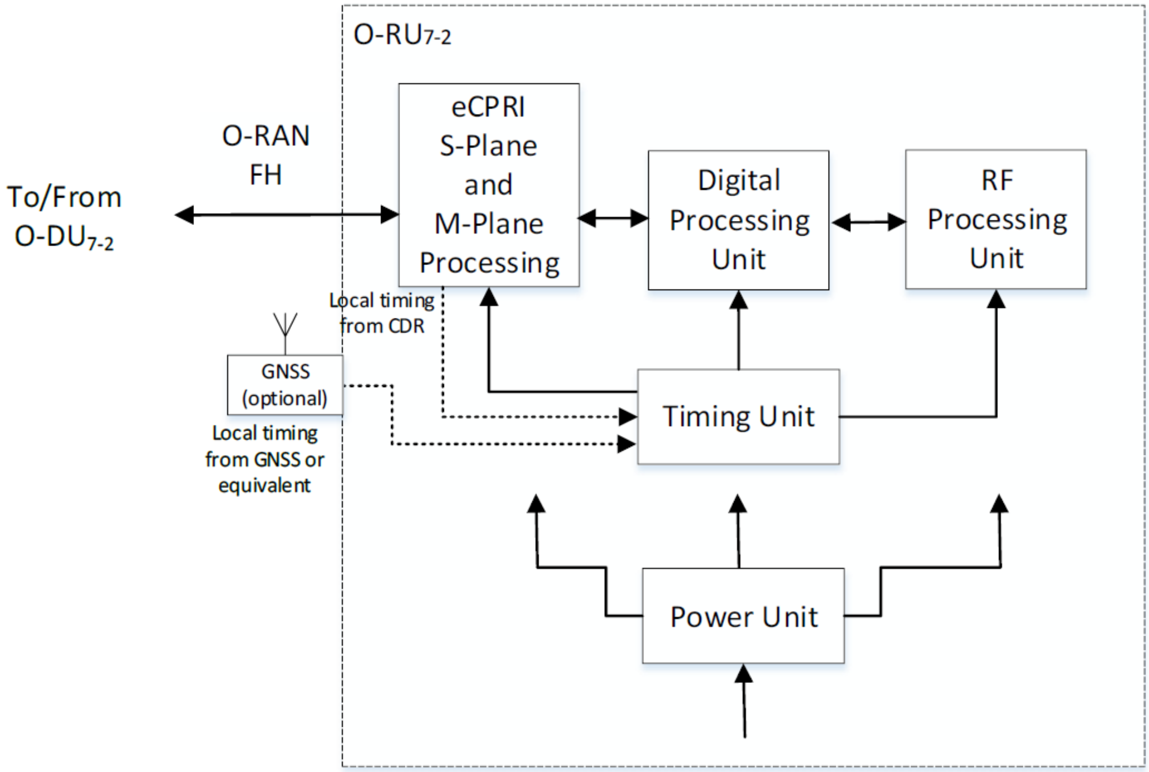

Figure 4-2: O-RU architecture

The NES techniques described herein can apply to FR1 and FR2 deployments. In general, the methods for energy savings can apply to both TDD and FDD. The FDD forms will be inherently different, however, and will be described in a separate section.

Cell type	OMAC, OMC, and EMC
Frequency Band	Any valid FR1 or FR2 band
Instantaneous Bandwidth	As documented in OMC, OMAC, and EMC HRDs
Occupied Bandwidth	As documented in OMC, OMAC, and EMC HRDs
Inter site distance	As documented in OMC, OMAC, and EMC HRDs
Antenna Configuration	4T4R ~ 64T64R
Fronthaul Type	O-RAN FH, Split Option 7.2x CAT-A and CAT-B Radio Units
Midhaul	Split Option 2

Table 4-1 O-RU Deployment options to which NES applies

4.2 Carrier On/Off Network Energy Savings

From WG7 point of view, carrier Off/On means disabling or enabling a tx/rx Array carrier. The feature aims at reducing O-RU power consumption by switching off/on one or more carriers of the radio access network.

Carrier switch off/on has two modes of operation: (i) Energy savings mode in which the radio's power amplifiers remain with minimum current draw, and (ii) complete switch off (M-Plane enabled).

In the frequency domain, there are at least three forms of carrier switch-off:

(a) Disabling all the component carriers (CCs) and RF bands used at the O-RU site.
(b) Disabling at least one of the multiple RF bands used at the O-RU site
(c) Disabling usage of one or more CCs within a band at the O-RU site.

All of the above actions shall allow for power reductions in the O-RU, with item (a) realizing the greatest reduction.

4.2.1 Disabling all the component carriers (CCs) and RF bands used at the O-RU site.

In this scenario, the cell or cell sector is disabled for some duration, e.g., several hours, and does not transmit or receive an RF signal for any of its supported Radio Access Technologies (RATs). It is still necessary for the O-RU to keep at least some of the fiber or wired links to the O-DU active, in order to receive M-Plane messages to resume operation, or report other management parameters.

During this O-RU_Full_Disable state, the actual power savings will be vendor-dependent. The figure below provides an exemplary depiction of which O-RU sub-blocks would be in the lowest power states.

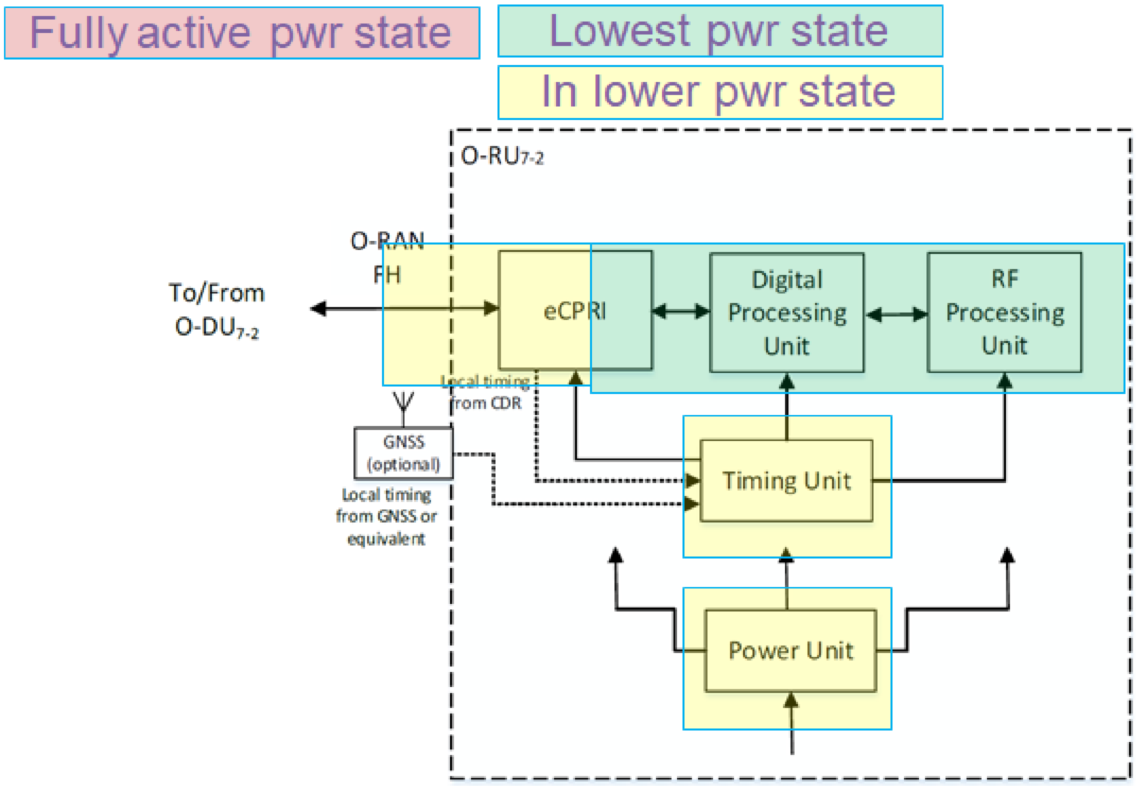

Figure 4-3: Energy Savings for O-RU Fully Disabled

The descriptors “Lowest Power State” and “Lower Power state” relate to the degree of functionality of each sub-block, which corresponds to the level of energy savings possible. Each sub-block will now be discussed:

(a) RF Processing Unit. Since the O-RU is in a state where all carriers for all RATs are disabled, no RF functionality is needed. Thus all RF blocks can be in the lowest power state. Specific supply rails from the Power Unit may potentially be disabled.

(b) Digital Processing Unit. In this context, the Digital Processing Unit relates to functionality directly associated with the RF signals, such as DFE, DPD, DDC/DUC, etc. Therefore, it can also be put into the lowest power state. Some supply rails may be disabled, or lowered in voltage to support retention of digital content.

(c) Power Unit. As mentioned above, some supply rails can be disabled, or reconfigured to a more efficient mode associated with lower load currents.

(d) Timing Unit. Power consumption may be lowered by disabling buffers or PLLs used to generate clocks for the RF Processing and Digital Processing Units. O-RU timing and frequency synchronization, as per IEEE 1588, should still be preserved.

(e) eCPRI Unit. C-Plane and U-Plane message handling functionality can be disabled, but M-Plane message processing support will still be needed.

(f) O-RAN FH. Some of the fiber link optical modules (e.g., QSFP) can be disabled since user plane data capacity is reduced to zero. The corresponding PHY transceivers that connect to the QSFPs can also be disabled. If instead wired Ethernet links are used instead of fiber, these unneeded Ethernet links can be put into one of the pre-defined Ethernet PHY low power modes.

The wake-up time for this O-RU_Full_Disable state would be the longest of any of the Carrier Disable modes, since all RF-related functionality is disabled; C-Plane message processing may also be disabled. The wake-up time needed to get at least some of the carriers back on the air is expected to take 10s or 100s of msec.

4.2.2 Disabling at least one of the multiple RF bands used at the O-RU site

In this O-RU_Band_Disable scenario, the RF functionality for at least one of the operational bands is disabled. RF traffic on the other operational bands would continue, but it is anticipated that overall user traffic would be less, as this would be the normal pre-condition for disabling one of the RF bands.

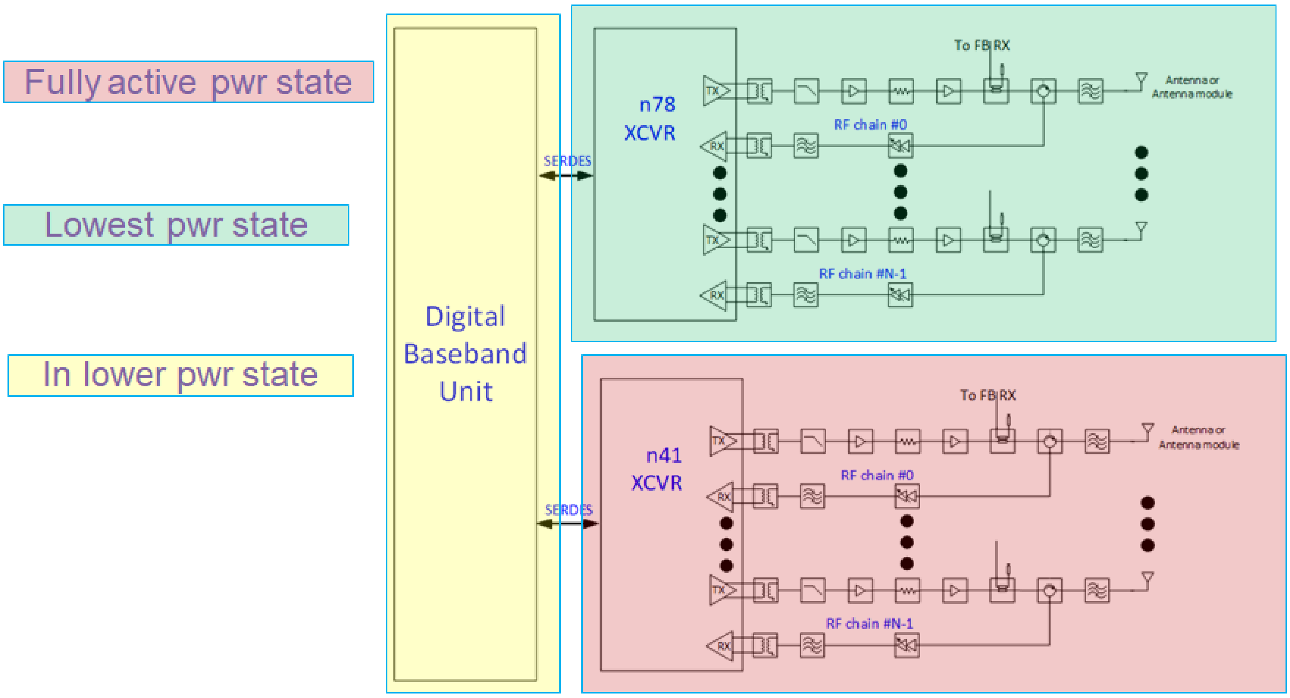

Figure 4-4: Energy Savings for O-RU Band Disable

The diagram above shows one example scenario where, for an O-RU supporting NR TDD bands n78 and n41, one of the bands (n78 in the diagram) is disabled during lower traffic scenario. The architecture shown has dedicated XCVRs for n78 and n41, but it could be one transceiver with only the n78 RF paths disabled.

In the figure, the n41 XCVR and associated RF front-end is shaded red to indicate full power operation. The n78 RF portion is shaded green to indicate it's in its lowest power state. The Digital

Baseband Unit is assumed to be one SOC or board that services both RF bands, and is shaded yellow to indicate it's processing less digital traffic. In another form, there could be separate Digital Baseband Units for each RF band, in which case the n41 Baseband would be shaded red, and the n78 one shaded green.

The recovery time to bring the disabled RF band back on the air is expected to be 10s of msec, but details depend on what sub-blocks were disabled in the RF XCVR. Data traffic can continue on the operational band, while the other band is being re-activated.

4.2.3 Disabling usage of one or more CCs within a band at the O-RU site.

In this O-RU_CC_disable scenario, the number of active component carriers (CCs) in a band are reduced, but the band still stays in operation. For example, as overall user traffic abates in the evening, the O-RU configuration could be modified to use two 50 MHz CCs instead of four.

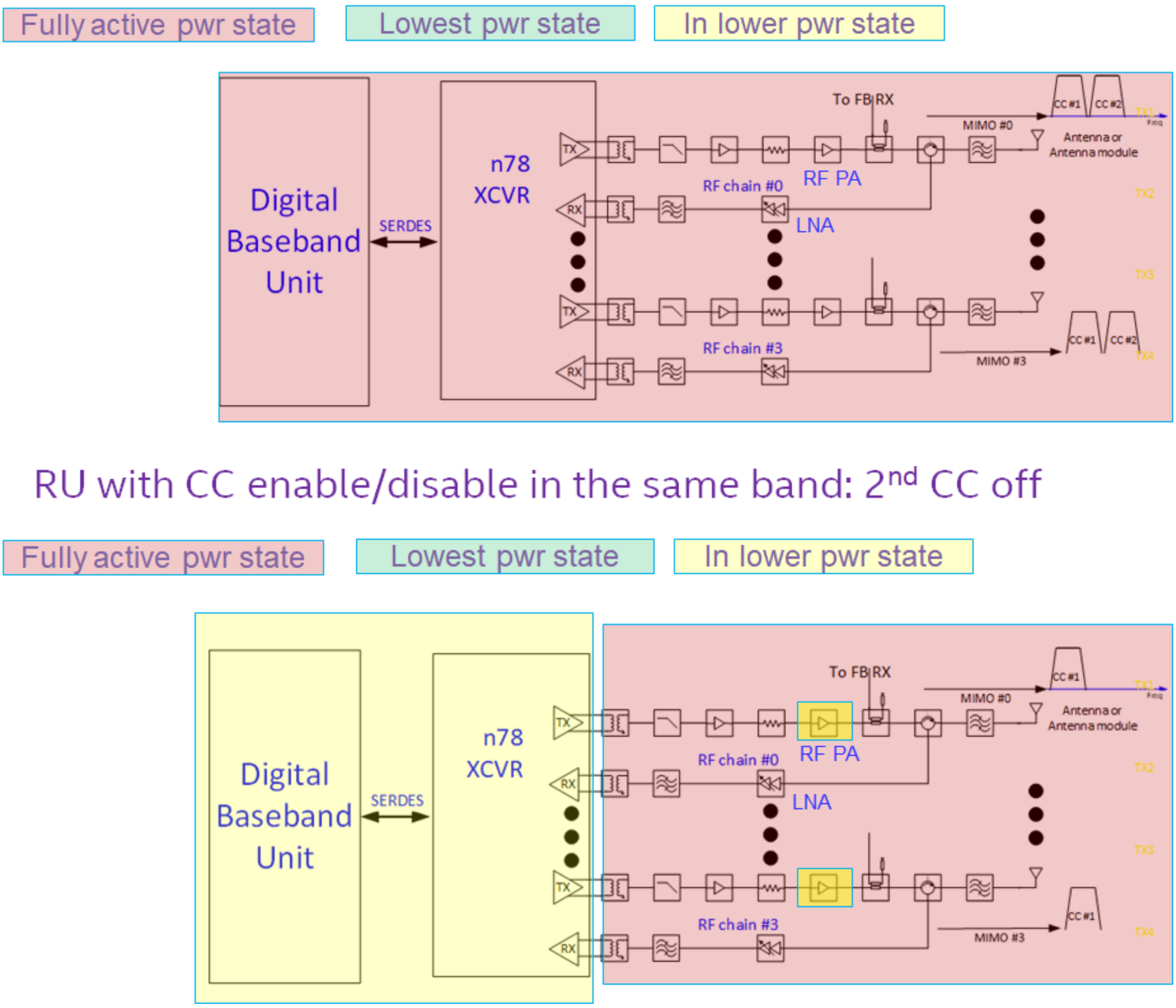

Figure 4-5: Energy Savings for O-RU CC Disable: two CCs active (top figure), and one CC active (bottom)

The two figures above show a case where two CCs are used initially, then the O-RU reconfigures to use only one CC when user traffic diminishes. In the top figure, all sub-blocks are operating at the nominal power level required for two CC operation. In the lower figure, the digital baseband and XCVR are in a lower power state, as fewer CCs are being processed. The actual power savings for the digital baseband and XCVR block depend on implementation details such as which sub-component performs CC combining and splitting, whether contiguous or non-contiguous CA is used, etc.

Note that, in the RF front end diagram in the lower portion, all sub-components are shown as red except for the PA. The other components will draw nearly the same power regardless of how many CCs are being sent. The RF PA, however, would be operating at approximately 3 dB lower output power (in the example above, and assuming full RB allocation for the two CC and one CC cases), so some reduction in energy consumption is expected.

The recovery time to re-activate the disabled CCs requires further study. A closely related aspect is whether data traffic can continue on the already active CCs while the other CCs are being re-enabled. If so, then more time can be allowed to re-activate the dormant CCs. The details will be implementation dependent.

4.3 RF Channel Reconfiguration Network Energy Savings

The scope of RF Channel reconfiguration includes following actions

(a) O-RU Array Rx/Tx ON/OFF e.g., 64Tx/Rx to 32Tx/Rx
(b) Modify SU/MU MIMO spatial streams or data layers.
(c) Modify number of SSB beams.
(d) Modify O-RU Antenna Transmit power

In mobile networks mMIMO radio unit has integrated antenna system that offers Single/Multiuser MIMO and beamforming capabilities to enhance cell capacity & throughput. The beamforming is achieved by exciting the array elements with appropriate weights. With this technique multiple uncorrelated beams can be generated, through which multiple data streams can be transmitted/received to/from UE during high-capacity requirement scenarios. In low-capacity scenario, i.e., when the expected traffic volume or number of connected users are lower than the configured threshold, energy savings can be achieved by reducing the power consumption of O-RUs by switching off certain Tx/Rx RF channels that are connected to the respective mMIMO array antenna elements/sub array. For example, 32 out of 64 RF channels connected to Tx/Rx Arrays of an O-RU can be switched off in a digital mMIMO architecture and correspondingly the number of spatial layers and SSBs can be reduced. The procedure (involvement of respective O-RAN interfaces) of the RF Channel Reconfiguration depends on the management architecture model (hybrid or hierarchical) and the deployment option. The off/on switch decision can be made by an AI/ML model within the inference host deployed at the non-RT RIC (denoted as deployment Option 1), or at the Near-RT RIC (denoted as deployment option 2) [11]. Among others the AI/ML models may include prediction of future traffic, user mobility, and resource usage and may also predict expected energy efficiency enhancements, resource usage, and network performance for different ES optimization states.

The main aim of RF Channel Reconfiguration is to perform O-RU Tx/Rx Array selection. However, Tx/Rx Array reselection may require modifying the maximum number of spatial streams, the number of SSB beams or the O-RU Antenna transmit power. Hence the overall scope of RF Channel Reconfiguration includes the actions listed in Table 4.3-1.

Action Name	Explanation	Possible Implementation Method
O-RU Tx/Rx Array selection	O-RU Tx/Rx Array selection means switching off certain Tx/Rx Arrays or Array elements to reduce power consumption of O-RU. Reselecting Rx/Tx Arrays may impact cell coverage.	− O-RU reports all supported Tx/Rx Array selections to O-DU or to SMO via Open FH M-Plane. − Based on traffic load and user distribution the Non-RT/Near RT RIC will optimize Tx/Rx Array selection.
Modification of number of SU/MU MIMO spatial streams or data layers	The maximum number of spatial streams may depend on the O-RU Tx/Rx Array selection, and it may be necessary to be modified along with a new Tx/Rx Array selection. A reduction of the number of spatial streams without Tx/Rx Array selection is not excluded as in certain scenarios the energy consumption of O-RU and/or O-DU may be reduced by turning off certain processing units of FPGA/GPUs, by running at a lower clock speed or turning off CPU/GPU cores along with reduced processing requirements.	− O-RU may report the maximum number of supported spatial streams for each Tx/Rx Array selection. For example, 16 spatial streams for 64 Tx/Rx and 8 spatial Layers for 32 Tx/Rx. − Based on the traffic load and the user distribution the O-RU will be instructed about the maximum number of spatial streams to be used. Hence, the O-RU and/or O-DU may be able turn off certain processing units that would be required to processing a higher number of spatial streams.
Modification of the number of SSB beams	The number of SSB beams may depend on the O-RU Tx/Rx Array selection and the number of spatial	− The number of SSB beams is controlled by the O-DU. Hence the O-DU can be instructed via Non-RT/Near-RT RIC to set the

	streams. Hence it may be necessary to adapt the number of SSB beams.	number of SSB beams based on Tx/Rx Array selection.
Modify the O-RU Antenna Transmit power	The maximum O-RU total transmit power may depend upon the O-RU Tx/Rx Array selection; hence it may need to be modified along with Tx/Rx Array selection to compensate for reduced coverage in the downlink.	− The O-RU power is configured per Tx Array. Hence, the modification may not be required when changing the O-RU Antenna selection from 64 Tx/Rx to 32 Tx/Rx. However, if possible, O-RU Tx power can be increased to compensate for reduced coverage.

Table 4-2: Some of the actions of RF Channel Reconfiguration [11]

4.4 Active Low Power Mode

Most of the Network Energy Savings mechanisms described so far lead to an explicit or implicit decrease in peak capacity due to the disablement of component carriers, MIMO/mMIMO data layers, number of antenna beams, and RF channel reconfiguration. These mechanisms are justified due to a decrease in network traffic or a change in the network operating environment. Another energy saving method i.e., “Active Low Power mode” can be employed by considering the reduction of the active mode power of the O-RU components, based on determining from network conditions and performance KPIs that such a change will not be deleterious to overall network performance. The rated O-RU traffic capacity would not change as all current active RF chains remain fully functional. Some examples of Active Low Power mode reductions are:

(a) Reduce RF transmitter chain power consumption at least in multiple Watts, at the expense of relaxed ACLR, based on determining that nearby channels’ performance would not be impacted by such a parametric change in the transmit chain.
(b) Reduce RF receiver front-end power consumption at least in hundreds of milli Watts, at the expense of relaxed front-end linearity (IP3), based on determining that average RX input levels have decreased. The average RX input levels can decrease due to reduced uplink traffic and/or decreased presence of “rogue” interfering signals.
(c) Reduce RF receiver power consumption, at the expense of relaxed adjacent channel selectivity (ACS), based on determining that RF blockers are not present in sufficient magnitude to warrant full ACS performance.

To determine if the RF chain power reductions have network performance impact, data exchange with the upper O-DU and CU layers may be needed. For example, the network would assess whether the O-RU power consumption reduction caused any degradation to the following KPIs, for both DL and UL traffic directions:

(a) HARQ retry rates
(b) Average MCS used
(c) Average throughput per RB sent
(d) Average energy per bit,

If it is determined that these metrics were not impacted in a statistically significant way, then the lower power O-RU mode could be considered safe for continued operation. If, however, a degradation in these metrics was observed, then the RF chain configuration would revert to the full performance/default mode, or an intermediate mode. The KPIs would be assessed independently for UL and DL directions, as the low power modes may be applied separately to the RX and TX signal flows. For an O-RU supporting multiple bands, these Active Low Power mode power reductions can be applied independently to each band.

The figure below shows the high-level functional block diagram of Active Low Power (ALP) mode implementation. Based on sensing RF and traffic metrics, the O-DU or CU would command the O-RU to enter a lower power mode for the RX or TX paths, or both. Performance data is periodically exchanged between O-RU and CU/DU to assess whether the power reductions negatively impact performance data. Some of the initial measurements to determine if the low power modes could be performed within the O-RU itself, e.g., average RX input power, incidence rate of signal overload events, etc. While normally the O-DU would command the O-RU to enter the Active Low Power mode, the O-RU could potentially engage this mode autonomously. This is certainly true in the case of an integrated O-RU/DU where the combined function box could autonomously determine network impact.

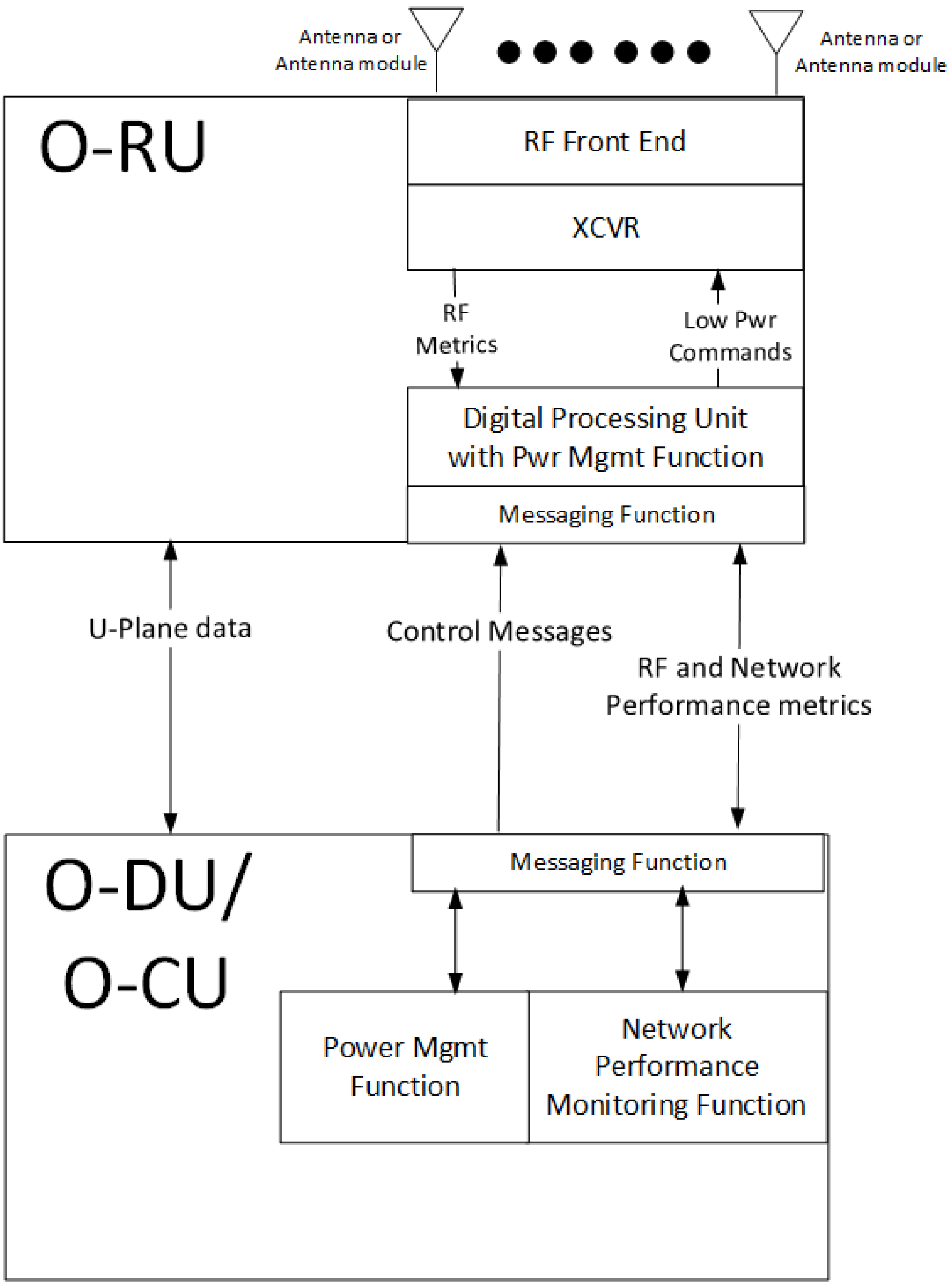

Figure 4-6 Active Low Power mode high level functional block diagram

4.4.1 Benefit and trade-offs:

(a) A significant benefit of Active Low Power mode is that, as the name implies, power is reduced while DL or UL traffic is active. Thus, the power reduction is not directly dependent on decreased user traffic.

(b) Another benefit is that the network conditions supporting the use of Active Low Power (ALP) mode would be expected to persist for several hours or even days. For example, late at night, the average user traffic would decrease, leading to statistically lower DL and UL RF levels being sent or received. On weekends, smaller "booster cells" associated with downtown business districts would be shut down or process much less traffic, potentially allowing ALP mode to be applied during the whole weekend.

(c) The decisions process and feedback mechanisms to support ALP mode are very amenable to AI and Machine Learning algorithms, as well as statistical hypothesis testing methods.

One drawback is that the procedure is somewhat empirical in that any network performance degradation cannot necessarily be predicted in advance. Thus, some short-term network impact could be seen when Active LP mode is first engaged, but the AI/ML "tuning" procedures could prevent this from happening in the long run. Additionally, network simulation models can also be used to predict the likelihood that any O-RU mode change would cause performance degradation.

4.5 TRX Control and Advanced Sleep modes (ASM)

In initial MVP-C proposals, Advanced Sleep modes were suggested as real-time messages to activate low power modes for durations on the order of multiple frames or slots. As per NES development in WG4, there are now four sleep modes, denoted as SM0, SM1, SM2, and SM3. An O-RU may support all of these sleep modes, a subset, or none at all.

The O-RAN CUS-plane spec, Rel. 13, incorporates new features and commands enabling TRX Control and Advanced Sleep modes. These were developed in WG4 NES Phase 2 sessions during the period January through July 2023. The released WG4 CUS-plane spec is available at

https://orandownloadsweb.azurewebsites.net/specifications

The NES ST4 (Section Type 4) commands are issued by the O-DU and processed by the O-RU; the ST8 (Section Type 8) command is sent in the opposite direction. The table below summarizes the salient features of these commands.

NES command-Section type	Cmd Scope	Sleep depth options	Sleep Duration	Wake-up time	Comment
TRX_control-ST4	ARRAY	SM0-SM3[1]	1 to (2^{28}-1) slots, or Indefinite[2]	1 to (2^{32} -1) usec for each of SM0-SM3[3]	
Advanced Sleep Mode (ASM)-ST4	ARRAY, CARRIER, O-RU	SM0-SM3[1]	1 to (2^{28}-1) slots, or	1 to (2^{32} -1) usec for	

			Indefinite[2]	each of SM0-SM3[3]	
Ready-ST8	Any of Above	Any of Above	Any of Above	Any of above	Required only if wake-up time is minimum rather than guaranteed

Table 4-2: WG4 CUS plane command summary table

1 Note 1: An O-RU may implement any subset of SM0-SM3, including all or none.

2 Note 2: An indefinite sleep duration is terminated by a subsequent wake-up command.

3 Note 3: Wake-up times are pre-defined, and read from o-ran-module-cap.yang. They can be guaranteed, or minimum. In the latter case, ST8 Ready message is used to tell O-DU that wake-up is complete.

A brief discussion of the WG4 NES changes, relative to the Rel. 12 CUS-plane spec, is provided below:

(A) In section 7.2.9.2.3 of the Rel. 13 CUS-plane spec, the TRX CONTROL Section Type 4 command is defined, with detailed description in sections 7.4.6 and 16.2. This command supports only ARRAY-COMMAND scope. It provides for a portion of the targeted antenna array to be put into one of the four sleep modes described below.
(B) In section 7.2.9.2.4, the Advanced Sleep Mode (ASM) Section Type 4 command is defined, with detailed description 7.4.6 and 16.3. The scope of an ASM command can be an antenna array (ARRAY-COMMAND scope), an array carrier (CARRIER-COMMAND scope), or the entire O-RU (O-RU-COMMAND scope).
(C) In section 16.1, a table of four sleep modes SM0-SM3 is shown. These command names apply to both TRX-CONTROL and ASM commands, but the wake-up times will vary depending on the target of the command. Below is the table included in v13 CUS plane spec, section 16.1.

	TRX CONTROL wake-up duration	ASM wake-up duration	NOTE
sleep mode 0	TRXC-mode0-wake-up-duration	ASM-mode0-wake-up-duration	
sleep mode 1	TRXC-mode1-wake-up-duration	ASM-mode1-wake-up-duration	L
sleep mode 2	TRXC-mode2-wake-up-duration	ASM-mode2-wake-up-duration	M
sleep mode 3	TRXC-mode3-wake-up-duration	ASM-mode3-wake-up-duration	N
NOTE: within clause 16 the shorter terms L, M, and N are sometimes used for these durations (either command)			

Table 4-3: WG4 CUS plane command Sleep mode wake-up duration table

a. It is required that the wake-up times comply with the following inequality: Sleep Mode 0 wake-up duration <L<M<N . Accordingly, it is expected that, for the same command object, ASM mode 3 saves the most power, and ASM mode 0 saves the least. (This is not a mandatory requirement, however.)

b. Note that, for any of the four sleep modes, applied as either a TRX_CONTROL or ASM command, the object of the command (array, carrier, or entire O-RU) will become non-functional and not be able to receive or transmit RF signals.
c. The amount of power saved in each sleep mode is O-RU implementation dependent.

(D) The Section Type 8 command "ready" field is used by the O-RU to indicate to the O-DU that the wake-up procedure is completed, in the case where the wake-up time is a minimum, rather than guaranteed. The minimum duration mode may apply for cases where the wake-up time may be extended by the need for perform a sub-system recalibration, e.g., if the temperature varied by a certain amount during the sleep interval.
(E) In section 16.12, an Emergency wake-up command is defined. This applies when a TRX_CONTROL or ASM sleep mode command is issued for a defined duration, but the O-DU later determines that it must wake-up the O-RU earlier. In this case, an M-Plane command is issued.

More details regarding the TRX CONTROL and ASM Sleep mode commands are provided in sections 16.6 through 16.13 of the latest CUS-plane specification. Some practical examples are given in section 16.11.

5 Network Energy Savings KPIs and Figures of Merit

In section 2.1, the listed references [5], [12], and [13] all describe KPIs relating to Network Energy Efficiency, and associated measurement methods. Generally speaking, the Energy Efficiency equation has the following general form:

Equation 5-1: Network_efficiency_KPI = (Network_parameter/Energy_Consumption)

where both Network_parameter and Energy_Consumption are measured over a prescribed time interval. The convention is that "larger is better", i.e., the Network_parameter should be as high as possible with Energy Consumption as low as possible.

Also, the scope of these energy metrics is typically the entire network, e.g., the gNB, backhaul, core network, and the link from the core network to the Internet or Cloud network. In this report, however, we will focus on KPIs pertaining to the O-RU, whenever possible. (Some KPIs may not be easy to measure at an O-RU level.)

5.1 Discussion of Energy Efficiency KPIs relevant to this NES report.

In section 5.3 of [13], three mobile network KPIs are discussed, relating to total user data volume, coverage area, and latency. The most direct measure is the KPI relating to data volume:

Equation 5-2: Data_energy_efficiency (DEE) = (Total_user_data_volume/Total_energy_Consumption)=(DV_{MN}/EC_{MN}) where

DV=User Data Volume (both UL and DL), EC=Energy Consumption, and MN= Mobile Network.

An O-RU-oriented version of this metric would measure only the O-RU energy consumption as well as the user data passing through the O-RU. While the O-RU energy consumption can easily be measured, it is not as straightforward to measure the O-RU data volume. This would require counting the PDSCH and PUSCH RBs sent, while also knowing the MCS used for each. HARQ retry RBs would also have to be accounted for.

It is possible to make some reasonable assumptions regarding the HARQ_retry RBs, as the MCS adjustment procedure in LTE and NR is designed to achieve a certain retry%, e.g., 10%. Regarding the MCS, with an integrated O-DU/O-RU deployment, the user downlink data volume is known since the O-DU function receives the actual user data units (PDUs). Based on these observations, we propose the following O-RU-centric Data Energy Efficiency metric:

Equation 5-3: Data_energy_efficiency_RU (DEE_{RU}) = (Total_num_PRBs/Total_energy_Consumption$_{RU}$)=($PRBV_{RU}$/EC_{RU}) where $PRBV_{RU}$ is the volume of PRBs sent or received by the O-RU in a certain observation interval, and EC_{RU} is the energy consumption of the O-RU in that interval. The correspondence between the PRB volume ($PRBV_{RU}$) and the actual user data will be the subject of further study.

Another KPI given in [13] is the Latency energy efficiency:

Equation 5-4: Latency_energy_efficiency (LEE) = 1/(Avg_end_to_end_latency*Total_energy_Consumption)=1/(T_{e2e} *EC_{MN}) where T_{e2e} is the average latency for user data packets traversing the network from the radio to the Internet. Note that both terms are in the denominator, since these KPIs follow the rule “larger is better”.

An O-RU-specific version of this metric is proposed:

Equation 5-5: Latency_energy_efficiency_RU (LEE_{RU}) = 1/(Avg_end_to_end_latency$_{RU}$*Total_energy_Consumption$_{_RU}$)=1/(T_{e2e_RU} * EC_{RU}) where T_{e2e_RU} is the average latency for user data packets flowing through the O-RU in a reference observation interval, and EC_{RU} is the energy consumption of the O-RU in that interval. The O-RU latency in the DL direction would be measured as the latency between when a user PRB arrives over the fiber link, in the form of an O-RAN User Plane data message, to the time that the corresponding RF I/Q data is sent out the TX antenna. Similarly, O-RU latency in the UL direction would be measured as the time between received RF I/Q samples at the antenna to corresponding PRB sent via User Plane message to the fiber.

Equation 5-6 DL Energy Efficiency = Total_RF_pwr_into_antennas/Total_O-RU_pwr_consumption

which would be measured during symbols where 100% of the RBs are transmitted. The KPI is adopted from the Open RAN MOU requirements [17] as discussed in section 5.2.3. The numerator quantity could be interpreted as the conducted RF power fed to the antenna port, for a BS Type 1-

C interface as described in section 4.3.1 of [3], or as radiated power measured at the Radiated Interface Boundary (BS Type 1-H as per 4.3.2 of [3]).

Measurement of the total RF pwr into the antennas cannot practically be performed during live operation. Therefore, it would either be:

(a) measured in a controlled laboratory environment, with the value stored in each O-RU's non-volatile memory, or

(b) declared as the rated RF power for the O-RU. A single, fixed number would be used for each O-RU, but it shall be consistent with lab measurements for a sample population of O-RU.

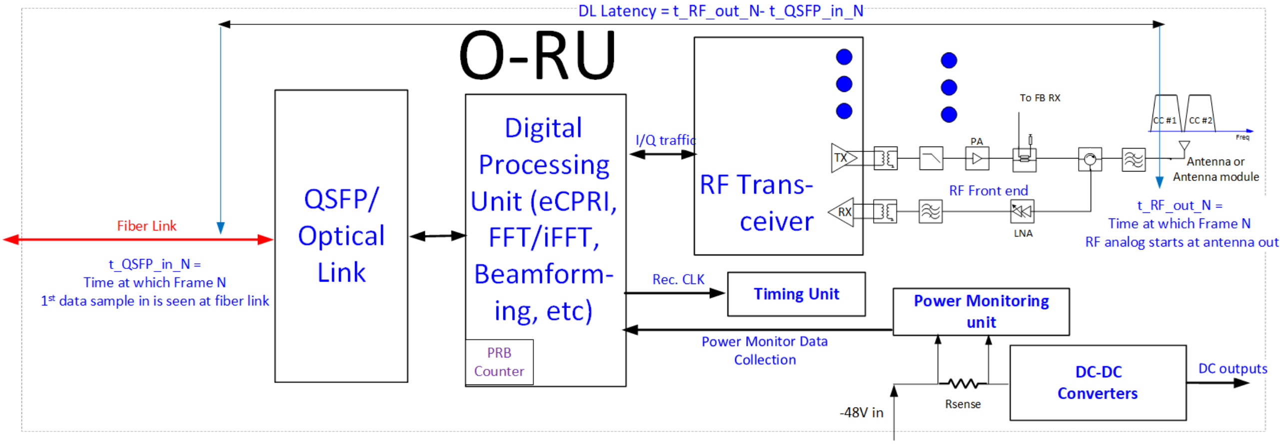

Figure 5-1: O-RU diagram showing metrics for E2E latency, PRB count, and power consumption

In Figure 5-1 above, the relevant O-RU metrics associated with the latency, PRB traffic count, and O-RU power consumption are shown. The figure shows a -48V powered O-RU but alternative sources such as AC or +12V supply inputs are possible. The O-RU power consumption can be measured at the -48V input using the power monitoring unit shown above. The DL and UL PRBs could also be monitored via PRB counters in the digital processing unit, or the O-DU could keep a record of allocated DL and UL PRBs. With an integrated O-DU, user data packets could be monitored, which will be a more direct measurement than PRBs.

While the end-to-end latency (shown above for DL traffic case) is easy to visualize, it may not be easy to measure in practice. More likely it would be characterized by design.

5.2 Discussion of O-RU-specific KPIs relevant to Network energy Savings

Here we provide a general discussion of the attributes or parameters associated with a given Network Energy saving mechanism. These may or may not impact the energy efficiency metrics discussed in the previous section.

5.2.1 Expected O-RU or O-RU/DU Network energy Savings Percentage

The basic expectation is that the Network Energy Savings methods described in section 4 all will reduce O-RU power consumption. The amount of savings will be implementation dependent, and O-RU or O-RU/DU vendors can differentiate their offerings by the degree of savings offered.

Even so, it is useful to state the amount of energy savings or power reduction expected by each method, as a percentage of some reference value of energy or power consumption.

The Carrier On/Off methods of Section 4.2 and the RF Channel Reconfiguration procedures of Section 4.3 will all reduce the Active mode reference power consumption of the O-RU, which is defined in section 3.1 for TDD mode as the average O-RU or O-RU+DU power for a 75/25 DL/UL symbol ratio, with no symbol gaps other than those needed for UE timing advance.

The following table summarizes the various Phase 1 Energy Savings mechanisms, and their impact on the O-RU sub-blocks. In the last column, a rough estimate of total O-RU % power savings is offered.

NES Mechanism	Traffic Capacity Impact	Wake-up Latency	Traffic Disruption during re-activation of disabled resources	% Energy Savings range vs. Active Mode Ref. Power Consumption	Comment
All RF carriers shutoff	No user Data traffic	100s of msec or several secs	N/A (already no traffic)	80-99%	
RF Band shutoff	Yes[1]		No	40-50%	Entire front end for disabled RF band can be put in very low power state
CC shutoff within a band	Yes[1]		FFS	Few %	Limited reduction in RF Front-end power; main power reduction will be via reducing JESD bandwidth and DUC/DDC filtering

RX/TX Array order reduction	Some impact[2]	msecs or 10s of msec	Likely	10s of %	Reducing number of RF front-end chains will directly reduce front-end power, which is dominant power consumer for medium and wide-area base stations
Reduced # of MIMO spatial layers	Likely			Unclear	Any change in # of antennas required could be offset by need for additional data symbols in time domain

Table 5-1 O-RU NES Mechanisms and their impact

Note 1: Traffic capacity impact should be directly related to the amount of RF spectrum removed from service.

Note 2: Traffic capacity impact depends on specifics of RF array re-configuration, e.g., reduction in # of beams could lead to reduced antenna gain, coverage range and DL/UL MCS utilized.

5.2.2 Diurnal O-RU power calculator

In references [5] and [15], methods for characterizing an NR FR1 base station power consumption in a defined set of test conditions are given. The base station activity model allows for three different loading states: (a) Low Load, (b) Medium Load, and (c) Busy Hour load. Additionally, as per Annex E of [5], user traffic is represented as a percentage of available PDSCH resources, and PUSCH is not explicitly accounted for. While these procedures are useful for measuring a candidate O-RU in a controlled lab environment, without having to generate simulated UE PRACH or PUSCH traffic, it may not fully represent the real-world network environment.

For actual network scenarios, we propose an O-RU diurnal energy calculator based on the actual Downlink, Uplink, and SM0-SM3 power state percentages. The general format is shown in the table below. As energy cost per kW-hr may vary versus time of day, it is also shown as a variable. The daily totals of interest will be total energy consumed, and total energy cost. Average time spent in each of the power states can also be computed. This power compute framework has multiple benefits:

A. Initially, during O-RU or network development, existing models of network traffic can be used to predict DL and UL percentages, with blank symbol periods then allocated to one of the four Advanced Sleep Modes SM0-SM3. Based on O-RU design targets for the different power states, total daily energy usage can be computed accordingly, along with trade-offs of network latency as a function of each sleep mode.

B. Later, in actual deployment, O-RU energy metrics can be captured in real time, along with the power state percentages1, and uploaded periodically to the upper network management functions. AI and Machine Learning techniques can be used to tune the network sleep mode behavior so as to minimize total energy cost, while still achieving acceptable network latency, user perceived throughput, and other network performance metrics.

Duration	DL %	UL %	SM0 %	SM1 %	SM2%	SM3%	Energy usage (kW-hr)	Energy Cost per kW-Hr	Total Cost
00:00-01:00									
01:00-02:00									
02:00-03:00									
03:00-04:00									
04:00-05:00									
05:00-06:00									
06:00-07:00									
07:00-08:00									
08:00-09:00									
09:00-10:00									
10:00-11:00									
11:00-12:00									
12:00-13:00									
13:00-14:00									
14:00-15:00									
15:00-16:00									
16:00-17:00									
17:00-18:00									
18:00-19:00									
20:00-21:00									
21:00-22:00									
22:00-23:00									
23:00-00:00									
TOTAL							Total daily energy		Total daily cost

The table above can be enhanced to include other refinements, such as DL and UL PRB utilization percentage for the used PDSCH and PUSCH symbols, PRACH and SRS symbol percentage, # of TRs activated, etc.

[1] The O-RU energy consumption would be reported by O-RU to O-DU periodically. The O-DU should know the percentages for each power state (DL, UL, SM0-DM3), so the O-RU need not track these.

5.2.3 Open-MOU Energy KPIs and requirements

The Open RAN MOU group, consisting of five European operators[2] , is not directly affiliated with the O-RAN Alliance. Its members are, however, part of O-RAN and have been working closely with the O-RAN Sustainability Focus Group (SuFG) to promote the Open RAN MOU requirements [1]. In particular, there are nine energy efficiency requirements (rel1_O-RU_26 through rel1_O-RU_34).

For example, rel1_O-RU_26 reads as follows "Energy Efficiency of RRH in 2T2R / 4T4R / 8T8R configurations shall be at least 40% (with 100% traffic load)". In discussions with SuFG, it came out that the definition is this KPI is really:

DL_O-RU_Energy_Efficiency = Total_RF_pwr_into_antennas/Total_O-RU_pwr_consumption

where the measurement is conducted with 100% of the total DL PRBs for the enabled carriers being transmitted. This KPI is now included in section 5.2.1, but O-RAN WG7 does not mandate any expected efficiency target.

5.2.4 O-RU Power calculation tool

WG7 has developed an Excel calculation tool to estimate O-RU power consumption, as well as the DL_Energy_Efficiency metric cited above. The tool will be located in the ZIP file containing this v2.0 technical report. Some snippets from the tool output are shown below. Some key points:

1. Normative Reference [16] provides some typical numbers for baseband and e-CPRI blocks, as well as PA output powers. Values similar to these are used in the O-RU Power calculation tool.

2. In the total output power roll-up in the right portion of the tool, for the TX power column, the DL energy efficiency power metric is computed.

3. While some of the detailed parameters such as PA efficiency, LNA power consumption, etc., are aligned with currently available RF Front end components, in no case were vendor power consumption values used literally.

4. The power for the baseband and e-CPRI components will depend on whether an FPGA or a dedicated ASIC is used, with the latter expected to draw less power.

Most importantly, the tool's purpose is to allow operators, RAN vendors, or SOC suppliers to estimate total O-RU power consumption and efficiency in different configurations. Each O-RU design will have a unique component set and thus different power consumption.

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[2] OpenRAN MoU Group - Telecom Infra Project

Parameter	Value	Units
TX_pwr_per_antenna_mMIMO	3	W
TX_pwr_per_antenna_Macro	30	W
Post_PA_filter_IL	1.5	dB
TX_pwr_per_antenna_dBm	34.8	dBm
FR1_pwr_per_PA	36.3	dBm
FR1_PA_eff	44%	(Module eff.)
FR1_PA_pwr_consumption	9.63	W
FR1_PA_disabled_pwr	0.06	W
Pre_driver_pwr_consumption	0.45	W
Pre_driver_Off_pwr_consumption	0.01	W
LNA_pwr_RX_active	0.45	W
LNA_pwr_RX_disabled	0.005	W
XCVR_pwr_per_RX_chain	0.95	W
XCVR_pwr_per_TX_chain	0.9	W
eCPRI_fronthaul_IC_pwr	35	W
L1_BB_DFE_IC_pwr	100	W
Timing_and_sync_module	2	W
QSFP_fiber_XCVR	15	W
UL_duty_cycle	25%	
DL_duty_cycle	75%	
PMU_SW_Eff	90%	
Energy_cost_per_kWh	0.16	
MIMO_Array_size	64	
UL_DL_ratio	0.333333333	
Total radiated power	52.8	dBm

Block or IC	**Nominal TX Pwr(W) for 64 Antennas**
RF XCVR	57.6
RX LNA	0.32
TX Pre-driver	28.8
RF PA	616.4
RF Sub-totals	**703.10**
eCPRI processing (FFT/iFFT/CUS plane)	35.0
L1 Baseband Processing	100.0
Timing_and_sync_module	2.0
QSFP_fiber_XCVR	15.0
Total_Pwr (W)	855.1
Total_Pwr_including DC_DC eff (W) for 64 TR array	**950.1**
DL_Efficiency (Pwr into antenna/Total O_RU pwr)	20.2%

6 Mapping of low power modes to M-Plane or C-Plane commands

The Network Energy Savings features described in Section 4 will ultimately be mapped to M-Plane or C-Plane commands in actual implementation. If the commands already exist, then this section will describe a general procedure for using the commands to realize the specific NES low power feature from Section 4. If the commands do not exist, then we discuss the basic requirements of the new command needed for effective NES feature implementation.

6.1 Feature to Command mapping table

The following table provides a high-level summary of the commands needed for each NES feature.

NES Feature	Document section	M-Plane or C-Plane control	Commands exist?	Comment/Follow-up
Disabling all RF bands	4.2.1	M-Plane	Yes	As per M-Plane section 15.3.2
Disabling one or more bands in an O-RU	4.2.2	M-Plane	Yes	As per M-Plane section 15.3.2
Disabling one or more CCs in an O-RU band	4.2.3	M-Plane	Yes	As per M-Plane section 15.3.2
RF Channel Reconfiguration	4.3	M-Plane, possibly C-Plane	No	WG4 discussion topic[1]
Active Low Power Mode	4.4	M-Plane	No	M-Plane command proposal pending[1]
Advanced Sleep Mode	4.5	C-Plane, M-Plane	No	Command formats now in WG4 discussion[1]

Table 6-1 NES features and mapping to M-Plane or C-Plane commands

Note 1: Specifically, as of this version 1.00 publication date, command formats for RF Channel Reconfiguration and Advanced Sleep Modes were being discussed in O-RAN WG4. The M-Plane command CR for Active Low Power mode is also pending as of this 1.00 publication date.

6.2 Detailed discussion of command mapping

Here we discuss the NES features listed above, and the M-Plane or C-Plane commands required to activate each NES mechanism. The discussion will be limited to providing general references to the M-Plane or C-command types involved, rather than outlining a very specific procedure. If in fact there are gaps, i.e., the necessary M-Plane or C-Plane commands do not exist, that is also noted.

Some of the capabilities referenced below may be optional O-RU features, e.g., the hardware-state feature. The required features will not be referenced explicitly in the NES capability discussion below, but the associated M-Plane or C-Plane sections should be consulted to identify any optional features or capabilities required to realize the NES functionality.

6.2.1 Disabling all RF bands

The first step is de-activating all the antenna carriers in the O-RU. This is already supported in the M-Plane and Yang model when all carriers, in all bands, have been placed in the inactive state. Section 15.3.2 of the latest M-Plane spec discusses carrier de-activation procedures. Then, via the NETCONF procedures described in M-Plane spec section 9.1.3, the O-RU Controller can place the O-RU into the global SLEEPING power state[3], where all O-RU functionality not needed for M-Plane support can be disabled. In this state, C-Plane commands to the O-RU cannot be guaranteed to produce a response, as the C-Plane command processor may itself be disabled.

To bring the O-RU out of the SLEEPING power state, the O-RU Controller can execute the NETCONF procedure to transition the O-RU into the AWAKE power state.

6.2.2 Disabling one or more bands within an O-RU

This amounts to de-activating all the antenna carriers for a specific RF band, while leaving other bands active. The O-RU is thus still capable of processing user data traffic. The NETCONF client can place all tx-array-carrier and rx-array-carrier elements in the band into either the INACTIVE or SLEEP state, as discussed in M-Plane spec 15.3.2.

6.2.3 Disabling one or more CCs in a band within an O-RU

This means de-activating one or more antenna carriers within an active RF band, while leaving other carriers within that band active. The O-RU is still capable of processing user data traffic

[3] See Yang module o-ran-hardware.yang

within that band, using the remaining CCs. The NETCONF client can place the unneeded tx-array-carrier and rx-array-carrier elements in the band into either the INACTIVE or SLEEP state, as discussed in M-Plane spec 15.3.2.

6.2.4 RF Channel Reconfiguration

RF Channel Reconfiguration amounts to de-activating groups of antenna carriers within an array, e.g., reconfiguring an rx or tx array from 64 elements to 32 elements. The purpose of such a reconfiguration may be to reduce the number of spatial streams/data layers, or reduce the number of antenna beams, or reduce the directivity of the existing beams.

The capabilities described in M-Plane spec 15.4.2 and 15.4.3 may be used here, but it is unclear if these are sufficient. O-RAN Work Group 4 is evaluating additional CRs that will facilitate RF Channel reconfiguration.

6.2.5 Active Low Power mode

Active Low Power mode is intended as a persistent low power mode that can be applied to the RX and/or TX sub-components within an active RF band, while RF receive or transmit is occurring. An M-Plane command will be used to enable it, but this command is still to be defined[4]. The scope of the command is expected to be all RX or TX chains within an array, but finer granularity may be possible.

6.2.6 Advanced Sleep mode

Advanced Sleep mode is intended to put RF and baseband processing chains within the O-RU into a low power state, for durations spanning one or more slots, sub-frames, or frames. The exact format of the Advanced Sleep mode commands is, as of this writing, now being developed in WG4.

Note that PRB and TX symbol blanking is already described in the C-Plane specification, sections 8.4.2.2 and 8.4.2.3. Additionally, section 7.7.7 SE7 describes how to aggregate multiple endpoints together for transmission blanking for Section type 0. Thus, multiple symbols for multiple endpoints (carriers) can be blanked using existing commands, but only within a slot.

7 O-RU Hardware changes required for NES support

The Network Energy Savings features described in this document may imply changes to the O-RU hardware architecture, and associated firmware, in order to provide an effective low power solution. In this Phase 1 TR, these are discussed at a high level.

[4] (As of the publication date of the 1.00 NES specification release)

7.1 RF Processing Unit changes

Referencing Figure 4-2, the RF Processing Unit may require the following changes to support significant power reductions:

(a) Ability to quickly disable and enable the front-end RF PA, PA driver, and LNA blocks. Typical control interfaces could be GPIO, SPI, I2C or MIPI RFFE.
(b) Ability to quickly disable and enable the RF transceiver chains that interface to the front end. The transceiver sub-components involved may include the ADCs and DACs, PLLs, mixers, channel filters, DFEs, I/Q interfaces[5], CPUs, etc.

7.2 Digital Processing Unit and Timing Unit changes

Referencing Figure 4-2, the Digital Processing Unit and the Timing Unit may require the following changes to support effective power reductions:

(a) Ability to quickly disable and enable the Digital Processing Unit sub-components. This may include I/Q interfaces[6], Beam Forming HW or SW functions, FFT/iFFT HW or SW functions, PRACH processing functions, O-RAN CU-plane processing functions, fiber interfaces, etc.
(b) Ability to quickly disable and enable the Timing Unit sub-components. This may include PLLs, clock distribution buffers, etc.

7.3 Energy Monitoring HW and data collection

The M-Plane data object epe-stats includes data fields for voltage, temperature, current, and power7. It can represent current/voltage/power for the entire O-RU or a sub-component. Figure 5-1 in section 5.1 shows the presence a power monitoring unit for measuring and recording the input power consumption. Some ICs that may be suitable for such a functionality are listed in Annex A.

Any power-monitoring chip suitable for O-RU will have an ADC capable of sampling input voltage and current, and computing power consumption. Additionally, digital averaging of voltage/current/power samples is typically provided.

[5] I/Q data formats could be based on JESD, UCIe, or other formats

[6] I/Q data formats could be based on JESD, UCIe, or other formats

[7] See M-Plane specification O-RAN.WG4.MP.0-R003-v12.00 or later, section 17.4 and Table B.1-1

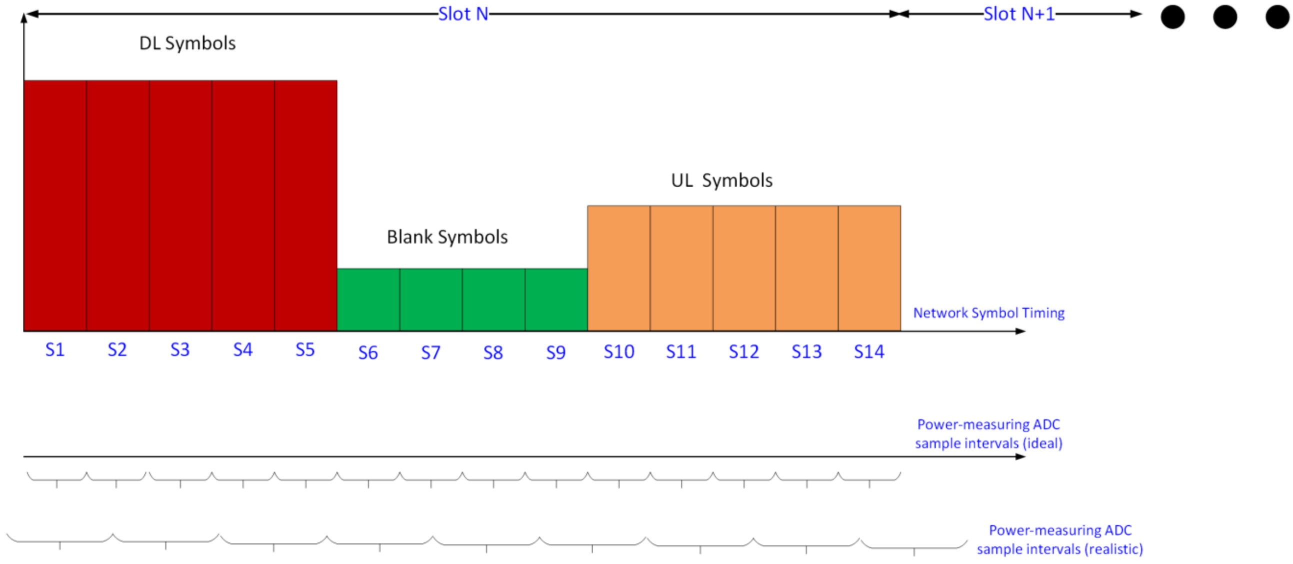

Figure 7-1: Power monitoring timing

The figure above shows an O-RU network traffic pattern example, with the power monitoring IC's ADC sample intervals shown. In the ideal case, we could have the ADC power sampling intervals be aligned in time with network traffic, producing a power reading per symbol, as shown in the second timeline. In practice, the power monitor's ADC timing may not be aligned with network timing, as shown in the bottom timeline, and the sampling intervals may be longer than one symbol or slot. Furthermore, the ADC sample timing may drift over temperature, if not synchronized with network clocks derived via SyncE protocols or other methods. To address this, coarser averaging can be performed, such that O-RU power data records spanning longer intervals, with fewer data points, can be uploaded to the O-DU. For example, a 1 Hz power sampling frequency can be considered, and reports uploaded every hour over M-Plane. If the amount of data per sample is 64 bits, this would result in an hourly data report size (excluding overhead) of 3600*64/ (1024*8) = 28.1 kBytes.

To avoid the impact of ADC sample drift, a power monitoring IC that accepts an external clock, derived from network timing, can be used. This ensures that a given number of ADC samples taken by the power monitoring IC always represents the same time interval.

Generally speaking, if epe-stats based monitoring is implemented in the O-RU, the following operating principles are recommended:

1. Power monitoring shall be supported for at least the O-RU input power feed.
2. The power monitoring IC sample clock should be synchronized to a clock related to Network timing.
3. No time gaps in the power monitoring shall be allowed.

7.3.1 Exemplary O-RU Power Delivery Network Architecture

To support O-RU NES monitoring and reporting capabilities, an example O-RU Power Delivery Network (PDN) is shown in Figure 7-2. The Telecom backplane of -48 Volts is used with Input Protection and Hot-swap capability for ease of insertion at the O-RU site. The Power Monitoring of current, voltage and power can be achieved by various methods, including DC-DC Converter, Switched Mode Power Supplies or Low Drop-out regulators with PGOOD (Power Good), IMON (Current Monitoring) and VMON (Voltage Monitoring) capabilities as well as power monitoring and sequencing ICs that individually control the sequencing of O-RU intermediate rails and their various power consumption metrics. The power monitoring information can be communicated back to the system digital processor via GPIO and low pin count communications buses such as PMBus or I2C/SPI. Additionally, support for the O-RU Alarm of a Dying Gasp event, is included in this example hardware design to keep the O-RU power alive long enough for the O-RU to communicate via M-plane that it has experienced a critical power failure. Please see Annex A for a list of possible part numbers to implement NES support.

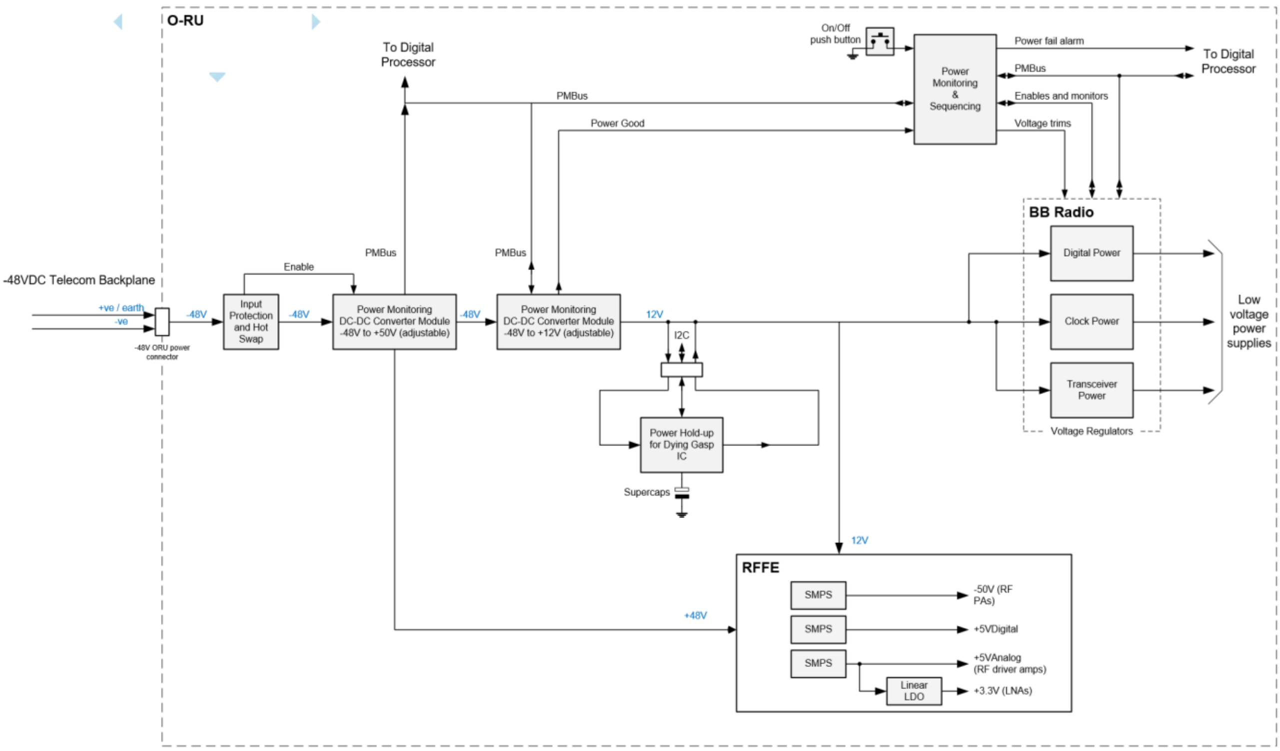

Figure 7-2: Exemplary O-RU Power Delivery Network support NES

7.3.2 Exemplary O-RU Temperature Sensor Architecture

To support environmental O-RU reporting for M-Plane data models such as epe-stats, an example temperature sensing architecture is shown in Figure 7-3. In this example, an eight channel temperature sensor IC with remote diode connected transistors is used to measure the temperature distributed throughout the O-RU in major subsystems such as digital processor, clocks, power delivery network, optical transceiver, DDR and other memories, RF transceiver, and RF Front-End components such as Power Amplifier and Low Noise Amplifiers. Similar to the power monitoring architecture, the temperature is reported back to the O-RU's digital processor via a low pin count communications bus such as SMbus, I2C, or SPI. Please see Annex A for a list of possible part numbers to implement NES support.

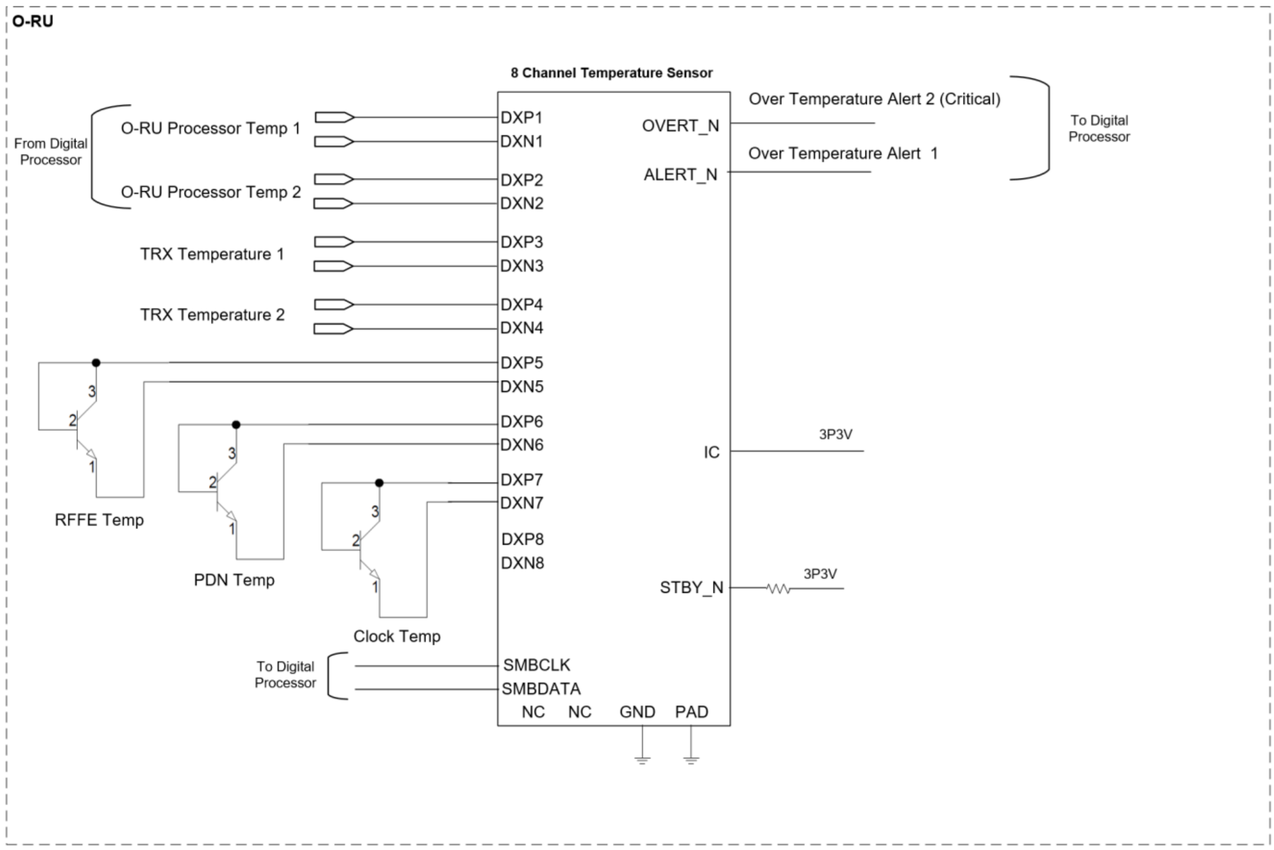

Figure 7-3: Exemplary O-RU Temperature Sensor Architecture supporting NES

7.4 Deep-hibernate

Deep-hibernate is an O-RU mode used to achieve relatively higher energy saving by disabling all carriers and stopping the C/U/S/M functions and corresponding processing units for a specified time duration. Deep-hibernate mode for the O-RU can enable a “Cell Switch Off” function where capacity cells can be shut down when not needed in an overlay environment within a coverage cell. Figure X shows an exemplary circuit to support the Deep Hibernate function of an O-RU. In this example, the objective of Deep Hibernate is to approach zero watts of power consumption since there are zero bits of data throughput required. Low power support circuitry is used to supervise the power rails of the O-RU as well as a real-time clock timer function that keeps track of the Deep Hibernate duration and wakes up O-RU components upon expiration. It should be noted there is no functioning M-plane to perform an emergency wakeup of the O-RU in Deep-hibernate mode. Upon expiration of the Deep-hibernate duration, the timer throws an alarm interrupt that the power sequencer uses to wake up the rest of the O-RU circuitry. It should be noted that this is only one exemplary implementation of how the O-RU can support the Deep Hibernate energy saving mode defined in the M-plane specification; there certainly are other supporting configurations that can achieve the same functionality with varying degrees of energy savings.

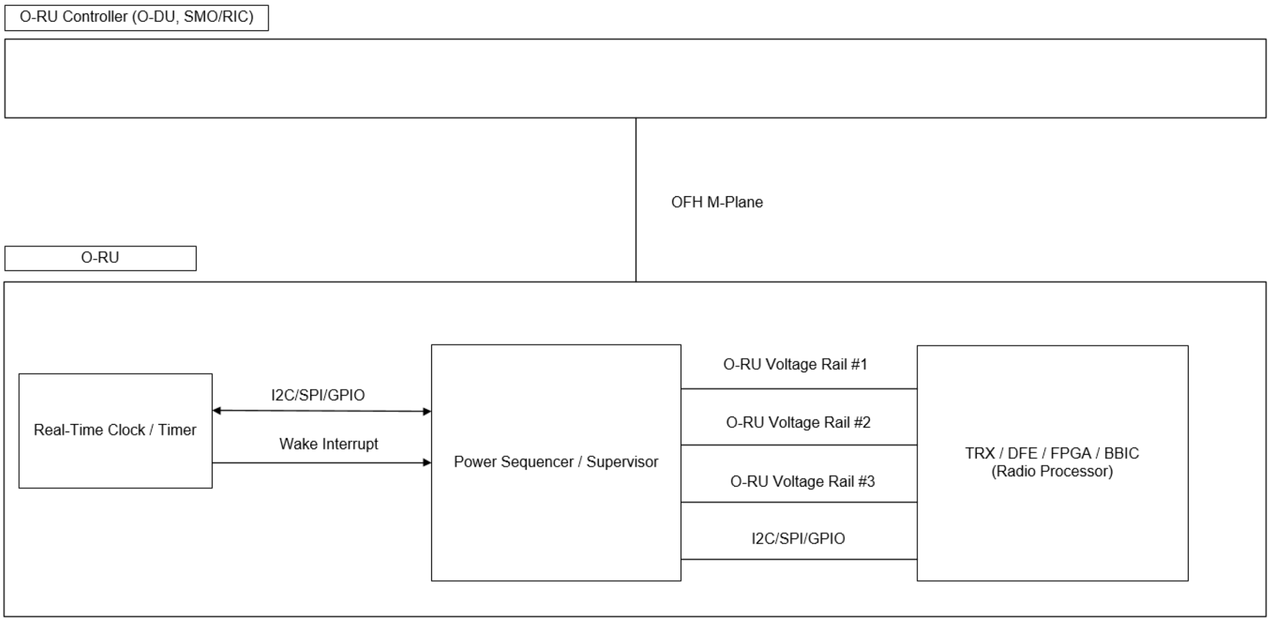

Figure 7-4: Exemplary Deep-hibernate support circuit. Power Supervisor/Sequencer and Real Time Clock with Hibernation Timers

7.4.1 Deep Hibernate implementation using SW timers

In the previous section, a HW-based mechanism is described to wake up the O-RU after the Deep Hibernate interval expires. Indeed, this approach can lead to the lowest power consumption as voltage regulators (VRs) for most sub-systems, including the Linux non-real time CPU complex, can be shut down, and re-enabled using the Power Sequencer component. The only clock required for controlled wake-up from Deep Hibernate is a low-frequency oscillator, e.g., 32.768 kHz RTC.

On the other hand, this ultra-low power implementation may lead to increased HW component cost. For example, the RTC IC and the advanced power sequencer IC shown in Appendix 8-2 may not otherwise be needed in the O-RU. Given the peak power of the O-RU will often be 100s of watts, a Deep Hibernate implementation which lowers power to a few 100s of mW may still be acceptable.

The figure below shows an O-RU block diagram with the Linux CPU cluster explicitly shown.

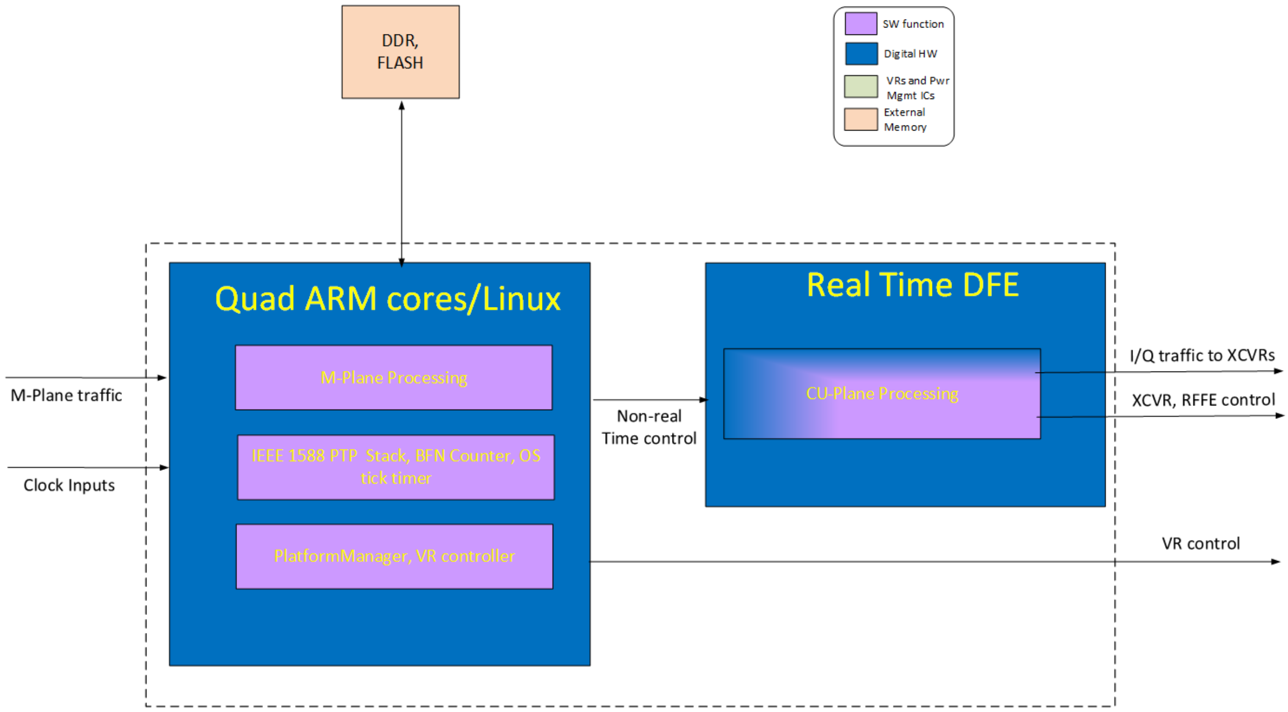

Figure 7-5: O-RU with Linux CPU cluster shown

The Deep Hibernate command is received via M-Plane and is intended to put the O-RU in a very low power mode for an extended duration, as indicated by the command pass parameter. In this SW-based implementation, the Linux CPU cluster would remain operational, but all other O-RU functions would be put in their lowest power state. Some exemplary pseudo-code procedures could be as follows:

(1) Received M-Plane Deep Hibernate Command
(2) Send commands to put real-time DFE, RF transceivers, RFFE, optical transceivers and SERDES in their lowest power stated.
(3) Shut off any VRs that can be disabled or put them in a lower power state (e.g., lower voltage or lower load handling capability).
(4) Configure Linux CPU cluster to a lower power state, e.g., for quad ARM core implementation, shut off three out of four cores.
(5) Configure Deep Hibernate wake-up task to re-activate after the Deep Hibernate interval has expired.
    a. This could be realized by an integrated HW timer, or by counting OS ticks. (Linux typically uses a 10 msec OS tick interval).
(6) When Deep Hibernate wake-up task re-activates, bring all subsystems out of low power mode.
(7) Inform O-DU or upper layers that the O-RU Deep Hibernate state has been exited.
(8) Resume processing of RF traffic.

In summary, we have described a Deep Hibernate implementation based on software procedures running on the non-real time CPU cluster (typically based on Linux). Since the Linux CPU must stay powered on, the power consumption may not be as low as a HW approach where VRs for all non-essential sub-systems, including the Linux CPU, can be shut off. It is still expected that Deep Hibernate power consumptions in the 100s of mW can be realized.

Another advantage of this approach is that a Linux reboot will not be required, leading to a faster Deep Hibernate wake-up time. Additionally, since the Linux OS is still running, other high priority interrupts can still be processed, though it is unclear whether we would expect any such interrupts in the Deep Hibernate state.

7.4.2 Comparison of SW and HW approaches to Deep Hibernate

OEMs should understand the tradeoffs associated with a HW approach vs using Linux SW to maintain timing. Generally speaking, with the SW approach, it is still necessary to keep a free-running VCXO or VCTXO operating in holdover mode, meaning that there is no input reference. Additionally, the Linux OS must keep running, albeit at a low duty cycle, which implies that its DDR memory is also active. All this translates to higher power for the SW-based approach. The table below summarizes some of the key tradeoffs.

Parameter	HW approach	SW approach	Comment
Pwr Consumption	Lowest power	100s of mW higher	Depends heavily on clock-synchronization ICs used, what clocks are maintained, whether Linux runs on SOC or FPGA, etc.
BOM impact	Some increase	No extra HW needed	HW approach may require an RTC clock and specialized VR sequencer
Re-boot time	Platform reboot may be required	No Linux reboot; DFE and XCVR may need re-initialization	S-Plane and PTP re-synchronization (required for both approaches) may take 10s of seconds; additional procedures such as VR restart, M-Plane re-initialization, etc., will further extend this.

Table 7-1 Comparison of HW vs SW approaches to Deep Hibernate.

7.5 Interfaces to Renewable energy apparatus

Solar-powered inverters are widely available now in the market, for both residential and commercial deployment. Well-known equipment manufacturers in this space include SMA, EnPhase, Fronius International GmbH, and Huawei. For an O-RU, we want to consider inverters with AC output power in the 1-10 kW range.

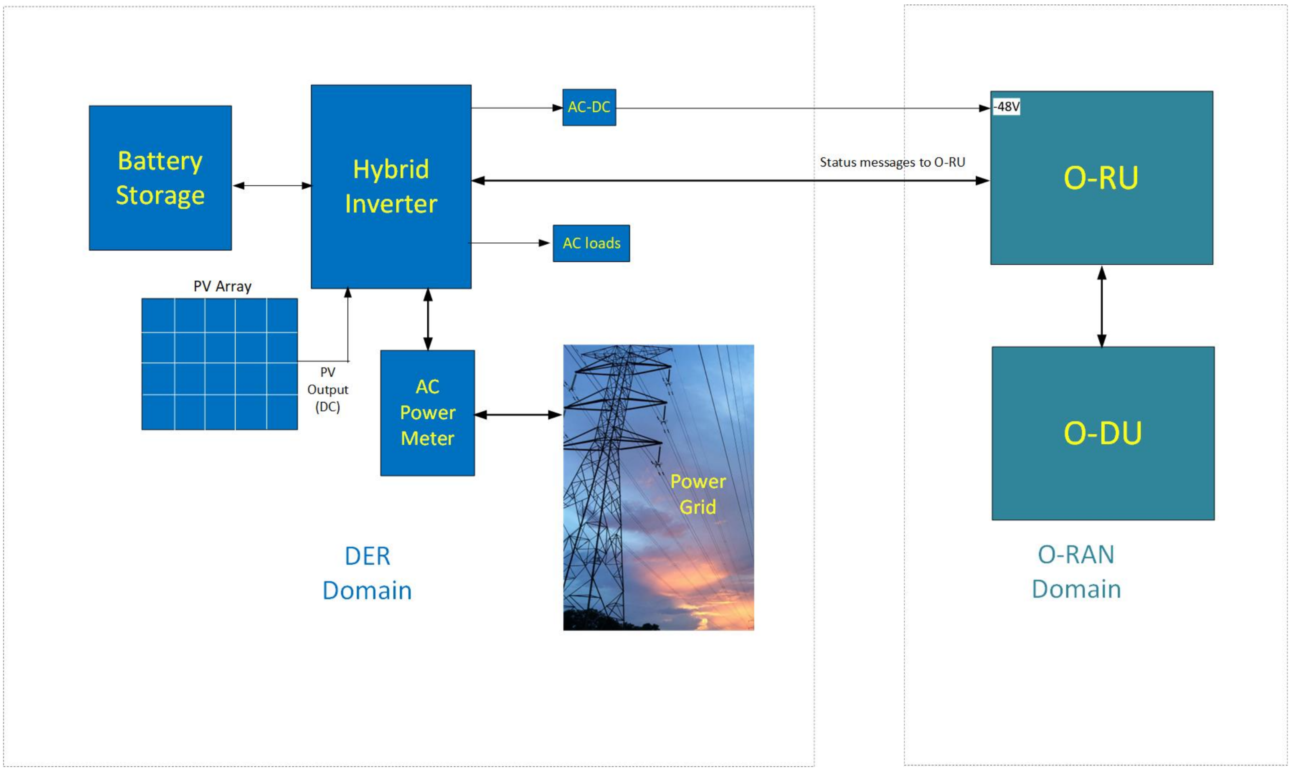

Figure 7-6: General Power System Block diagram with Renewable Energy Sources and Storage

The generalized distributed energy resource (DER) system diagram for a deployment providing solar power, with battery storage, to an O-RU, is shown above. The hybrid inverter would normally generate 110/220V AC power from either a PV (solar panel) source, battery storage, or the AC grid itself. The inverter will also control the charging and discharging operation of the storage batteries. In the case of an O-RU, it is common to use -48V as the input power source. As this would normally be generated from the AC grid, we can expect a similar approach to be employed when renewable sources are present and the figure above depicts this. Other options could be to derive -48V directly from the solar panel output or the storage batteries.

7.5.1 Solar Power inverter interfaces and protocols

The interfaces discussed here have two facets: physical/PHY level interfaces, and the accompanying command protocols. Generally speaking, the solar inverters targeting residential deployment will offer only Ethernet, Wi-Fi, and cellular connectivity, whereas those focusing on the commercial sector may also offer RS-485. There does appear to be at least one standardized protocol (Sunspec[8]) for exchanging appropriate command and status information between Distributed Energy Resources (DER). In an O-RU, which may only offer a Linux OS, appropriate Sunspec APIs would need to be installed. Thus, if the solar inverter supports SunSpec, the O-RU can in principle access the appropriate status info for the renewable energy resources. There is at least one organization offering objects and applications for Python language development[9] in a Linux environment.

In addition to SunSpec, other protocols referenced in [3] of Section 2.2 include (a) IEEE Std 2030.5 (SEP2), (b) IEEE Std 1815 (DNP3), often used in SCADA systems, and (c) IEC 61850, which describes information models related to DER devices, along with Internet protocols for data exchange between devices.

Sunspec describes an information model framework to describe DER devices such as inverters, photo-voltaic sources (i.e., solar panel array), storage sub-systems, environmental sensors, etc. The model fields include information such as component model #/serial number, support for standard SunSpec protocol fields, and vendor-specific fields.

In Annex A, Table 8-3, we show different interfaces and protocols offered by companies with products in the residential and commercial solar inverter ecosystems.

7.5.2 M-Plane messages to describe renewable energy system status

This is a subject for future study in WG4.

7.6 Dynamic PA Bias Control

RF PAs for O-RUs typically fall into two categories: lower power modules used for massive-MIMO arrays, and higher power PAs for macro base stations. The lower power modules may typically produce 34-35 dBm average power, and operate from a 28V rail, while the macro PA modules may output 40-42 dBm average and require a 48V supply rail for the output stage. There are several known methods for dynamically adjusting PA operating modes, including bias control. These are now discussed below.

[8] www.sunspec.org

[9] https://pysunspec.readthedocs.io/en/latest/pysunspec.html

7.6.1 PA On/Off Control

Most of the PAs available for O-RUs will offer a PA On/Off control signal, such that PA power consumption is greatly reduced without having to remove the main driver supply. This is useful for blank TX symbols where no data is transmitted. For effective operation, the PA On time should be 1.0 usec or less, and the total Idle power consumption should be in the 10-30 mW range per PA.

7.6.2 Dynamic Bias current control

In some literature [5], methods involving adjustment of PA bias current are described, e.g., for bipolar devices. For GaN technology base station PAs, this is not considered to be as effective as supply voltage adjustments described in the next two sections.

Dynamic Power Amplifier (PA) Bias Control is an energy-saving technique employed to reduce power consumption in O-RUs, as the PA is a significant power-consuming component. This feature, implemented through vendor-specific methods, involves the O-RU tightly controlling the PA bias via its Digital Front-End (DFE) and Crest Factor Reduction (CFR) functionalities. By adjusting the bias, the O-RU can optimize the PA's operating region, leading to reduced power consumption. The specific implementation details are dependent on the O-RU's hardware reference architecture.

The O-DU can also play a crucial role in managing PA power consumption. Knowing the Physical Resource Block (PRB) scheduling and corresponding maximum power requirements, the O-DU can leverage the Maximum Transmit Power Control feature defined in Clause 20.6.1 of the WG4 M-Plane specification [xx]. This energy-saving method allows the O-DU to restrict the number of scheduled PRBs, enabling the O-RU to adapt the PA's power output accordingly. By limiting PRBs, the O-DU effectively signals to the O-RU that it can reduce PA power.

7.6.3 Average Power Tracking

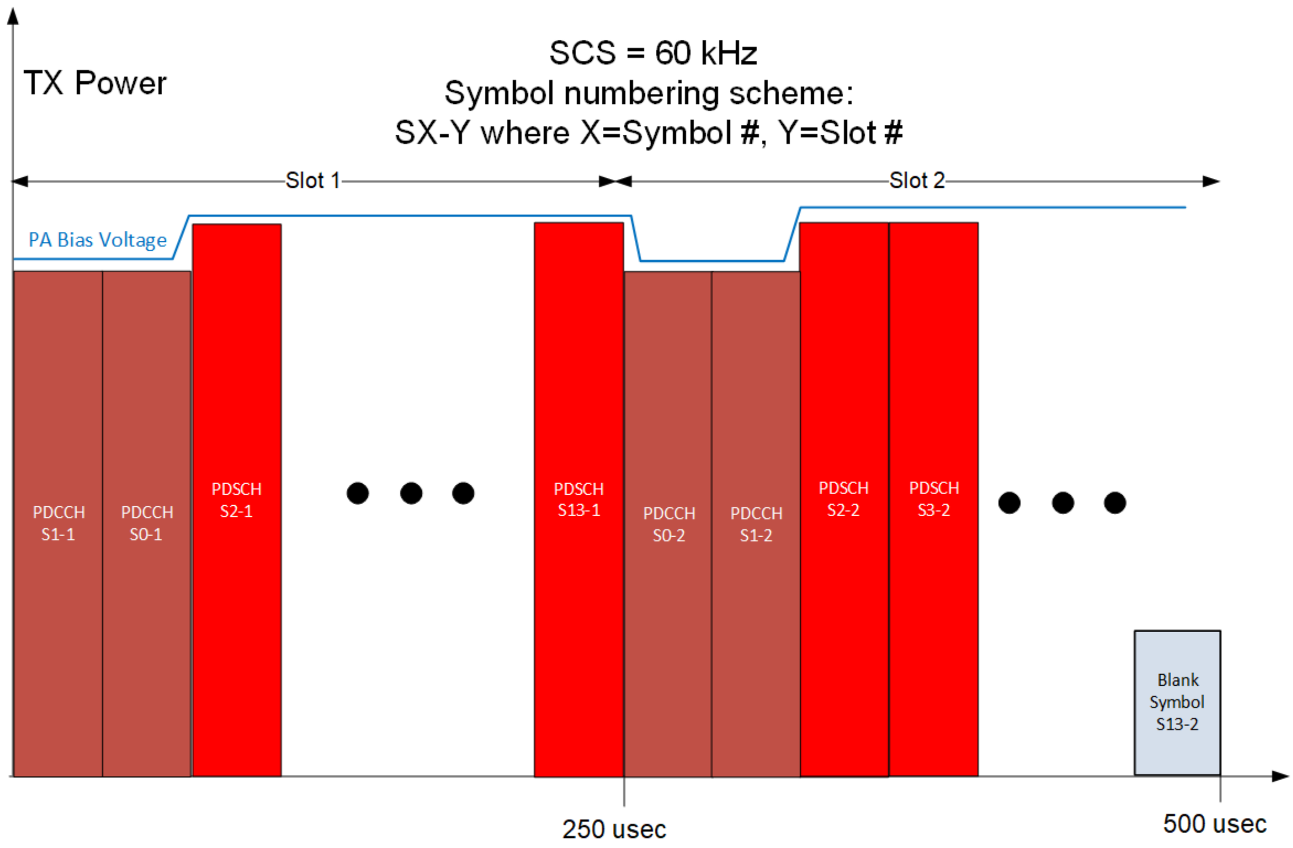

Figure 7-7: TDD DL slot example with blank symbols

In LTE and NR, the power per symbol is not guaranteed to be the same for all symbols in a slot. The diagram above shows two slots in a TDD configuration[10] having two PDCCH symbols transmitted at the beginning of the slot, followed by twelve PDSCH symbols in the first slot. In the second slot, the pattern repeats except that some of the symbols at the end could be blank, if overall traffic is light. In the general case, the PDSCH and PDCCH power levels will be different. This suggests that the PA efficiency could be improved by lowering the PA Bias voltage for the symbols having lower power (PDCCH symbols in exemplary diagram shown above.) In the case of the blank symbols, the PA will be completely disabled as discussed in 7.6.1; the bias voltage would not necessarily have to be reduced.

Alternatively, the PA supply voltage could be adapted at the beginning of every slot, to avoid the frequent transient settling times associated with changing it every symbol.

[10] These could be the first two slots in a DL/DL/Mixed/UL slot format.

7.6.4 Envelope Tracking

Envelope tracking techniques offer the best power savings, as the PA power supply bias voltage is varied real-time to track the power envelope. There is some existing literature in [6], [7] discussing envelope tracking implementations with GaN devices for base stations. The primary challenges are (a) supporting instantaneous modulation bandwidths of 100 MHz or greater for NR, (b) accounting for DPD, (c) accommodating the large amounts of bulk capacitance typically present on PA Vdd supplies, and (d) complying with 38.104 ACLR specifications.

7.6.5 Dynamic PA Bias Control Summary

The applicability and efficacy of all these techniques depends heavily on the O-DU scheduler algorithm, which could choose to repeat user data symbols for more S/N margin, spread out user data over more symbols for time diversity gains, etc. As scheduler operation is proprietary, we cannot make general conclusions about a specific RAN vendor's approach.

Additionally, in WG7, we reached out to various PA vendors and asked them to share any specific PA dynamic bias techniques that they felt could differentiate their offerings, as well as illustrate methods relevant to LTE and NR. Unfortunately, no PA vendors actually came forth with material to share with WG7.

8 Annex A: NES-related components for Power Monitoring

The table below lists some exemplary components that can be considered for O-RU power monitoring. The list is not intended to be complete, and equivalent ICs from other suppliers are likely available.

P/N	Description	Comment
Analog Devices ADM1075	-48V Power Monitor IC with Hot Swap support	5 kHz sample rate for input voltage and current; up to 128 readings can be averaged
Maxim 78M6613	Single Phase AC Power Monitor IC	Measures or computes AC voltage, current, line freq, real and reactive power

Analog Devices ADM1266	Cascadable Super Sequencer with Margin Control and Fault Recording	Power supply supervision and sequencing up to 17 supplies.
Analog Devices LTC3351	Hot Swappable Supercapacitor Charger, Backup Controller and System Monitor	Used to keep O-RU power up while Dying Gasp issued to M-plane.
Analog Devices LTC2980	Sequence, Trim, Margin, and Supervise Power Supplies	Multi-Channel PMBUS Power System Manager.
Flex BMR457 Series	Fully regulated Advanced Bus Converters	Measures and Monitors Current, Voltage, and Power.
Analog Devices MAX6581	8 Channel Temperature Sensor	Temperature Management & Alerts via SMBus.

Table 8-1: Catalog parts for input power monitoring

Additional P/Ns will be added if they provide differentiating features.

The table below lists some exemplary components that can be considered for O-RU Deep Hibernate support per Figure 7-4. The list is not intended to be complete, and equivalent ICs from other suppliers are likely available.

Component #	P/N	Description	Comment
1	Analog Devices ADM1266	Cascadable Super Sequencer with Margin Control and Fault Recording	Power supply supervision and sequencing up to 17 supplies.
2	Analog Devices DS1339C	I2C Serial Real-Time Clock	Manages O-RU Time Keeping functions including setting of alarms for NES Modes such as Deep Sleep and Deep Hibernate functions amongst others.

Table 8-2: Catalog parts for Deep Hibernate support

Annex A continuation

Supplier	HW interfaces	Command protocols	Comment
SMA	Ethernet, WLAN, RS-485, BAT-CAN	SunSpec Modbus, Speedwire /Webconnect, SMA Battery Interface	Some of these protocols are SMA proprietary
Enphase	Wi-Fi, LTE		
Huawei	RS485, WLAN/Ethernet smart dongle, LTE Smart dongle	EMMA	EMMA=Energy Management assistant box
Tesla Powerwall	Ethernet, Wi-Fi, LTE	Tesla Mobile App	Focus on Residential deployment
Fronius International GmbH			

Table 8 3: Solar Inverter suppliers and supported Interfaces.

9 Revision history

Date	Revision	Description
2023.06.13	01.00	Initial release of v1.0 NES specification
2024.07.16	02.00	V2.0. Candidate submission
2025.07.1	02.8	Interim version with nearly all CRs incorporated; added sections 7.4 (Deep Hibernate), 7.5 (Renewable Energy Sources), 7.6 Dynamic PA Bias Control
2025.07.8	2.9	Updated Interim version with nearly all CRs incorporated; added sections 7.4 (Deep Hibernate), 7.5 (Renewable Energy Sources), 7.6 Dynamic PA Bias Control
2025.07.15	3.0	Final Candidate submission